



Service Bulletin Number: 3810387

Released Date: 02-nov-2015

Analysis and Prevention of Bearing Failures

Analysis and Prevention of Bearing Failures

Contents

- Section 1 - Bearing Introduction
 - Nomenclature
 - What is a Micron?
 - Construction
 - Design
 - Bearing Surfaces
 - Bearing Lubrication
 - Bearing Load
 - Metal and Material Properties
- Section 2 - Bearing Failure and Diagnostic Process
 - What Causes Bearings to Fail?
 - Diagnostic Procedures
- Section 3 - Bearing Damage Patterns and Reuse Guidelines
 - Wiping
 - Loss of Overlay
 - Deep Wear
 - Overlay Fatigue
 - Lining Fatigue - Trimetal Bearing
 - Lining Fatigue - Bimetal Bearing
 - Corrosion
 - Lack of Lubrication
 - Cold Start/Fast Start Main Bearing Seizure
 - Geometric Deficiencies
 - Incorrect Connecting Rod Bolt Clamp Load
 - Assembly Errors - Main and Rod Bearing Caps
 - Assembly Errors - Main and Rod Bearings
 - Debris Between the Bearing and Bore
 - Loss of Bearing Crush
 - Cavitation Erosion
 - Electrical Current Damage
 - Debris Damage
 - Harmonic Lobing Break-in
 -

- Section 4 - Prevention of Bearing and Bushing Failures
 - Cleaning and Preparation - Cylinder Block
 - Cleaning and Preparation - Crankshaft
 - Cleaning and Preparation - Connecting Rods
 - Assembly - Main Bearings and Crankshaft
 - Assembling Connecting Rods to Crankshaft

Introduction

This Service Bulletin provides information on the bearings and bushings used in engines manufactured by Cummins Inc. The purpose of this Service Bulletin is to provide the user with better understanding of bearing and bushing design, construction, and functional requirements. With this better understanding, the user is better equipped to analyze bearing and bushing failures and how to prevent their occurrence.

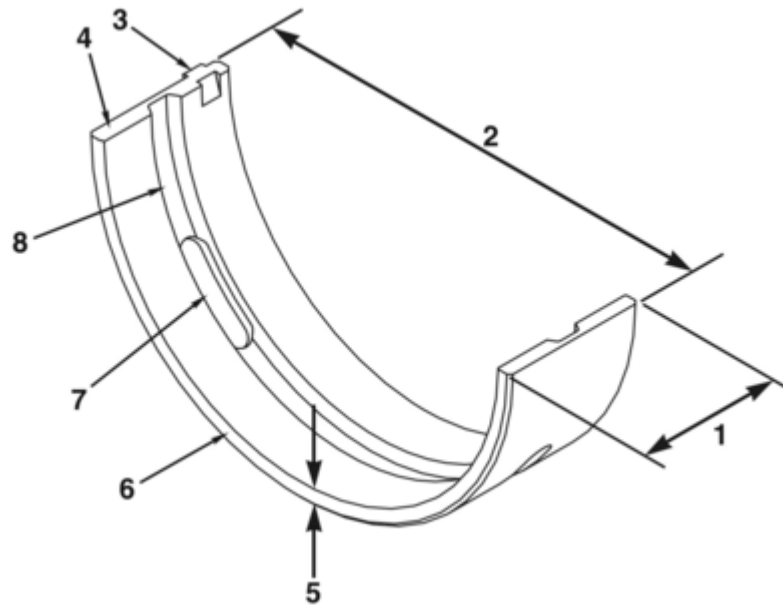
Substantial gains have been made in the design and material for shaft bushings and bearing in recent years. More work is underway to design and introduce bearings with linings containing little to no lead to protect the environment during the production, use, and disposal of bearings and bushings. However, acceptable bearing life is still dependent on correct assembly, adequate lubrication and proper oil and oil filter maintenance. When any of these conditions are unsatisfactory, the parts can be damaged in distinctive patterns. If the damage is caught in the initial stages it can be analyzed to determine which of the conditions was present and most likely responsible for the failure.

More importantly, the exercise of analyzing bearing and bushing damage will increase the understanding and appreciation for the conditions causing bearing damage so that preventive efforts are more logical and justifiable.

This publication also provides guidelines and guidance on a type process to investigate an engine problem that might be the failure of a bearing or bushing.

Nomenclature

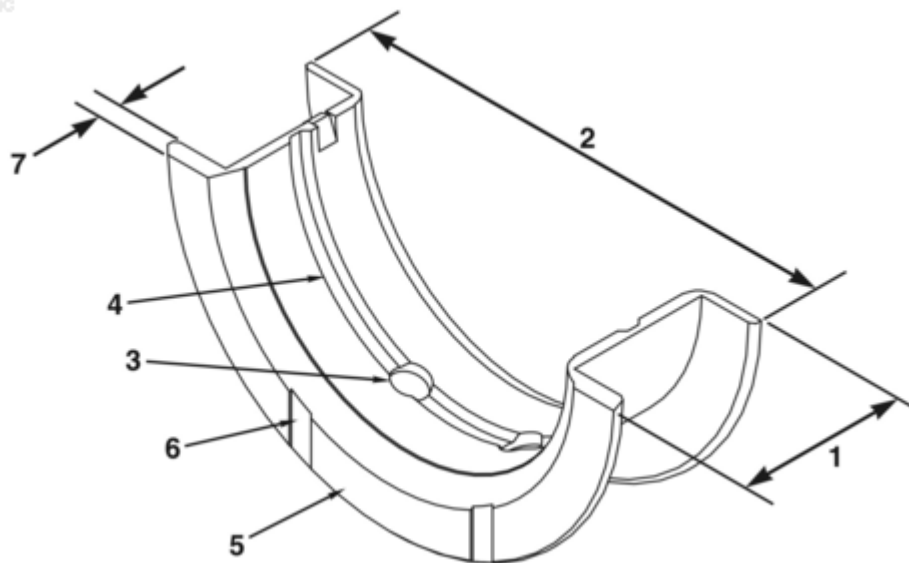
Knowledge of the names and descriptions of bearings will help to analyze and communicate the findings.



01r00072

Figure 1: Straight-shell bearing nomenclature (upper crankshaft bearing).

1. Bearing width
2. Bearing outside diameter (free spread)
3. Locating feature (tank, lug, nick)
4. Joint face
5. Wall thickness
6. End
7. Oil hole
8. Oil groove.



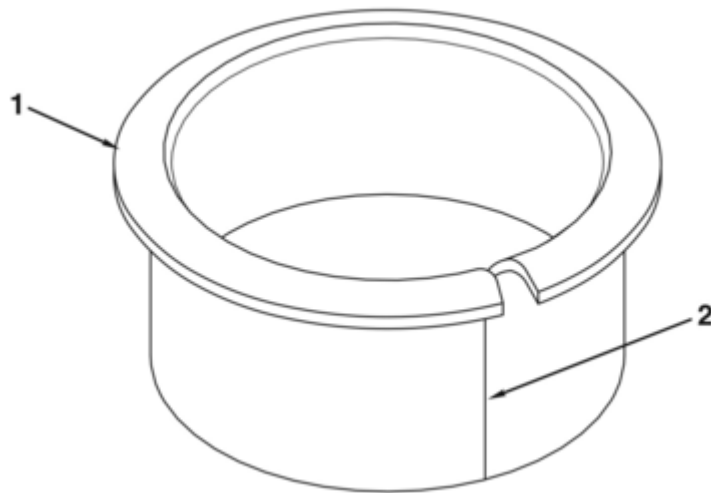
01r00073

Figure 2: Shell bearing with crankshaft thrust flange (upper crankshaft bearing).

1. Bearing width (free spread)
2. Flange outside diameter
3. Oil hole
4. Oil groove

- 5. Flange (thrust face)
- 6. Flange oil groove
- 7. Flange thickness.

©Cummins Inc

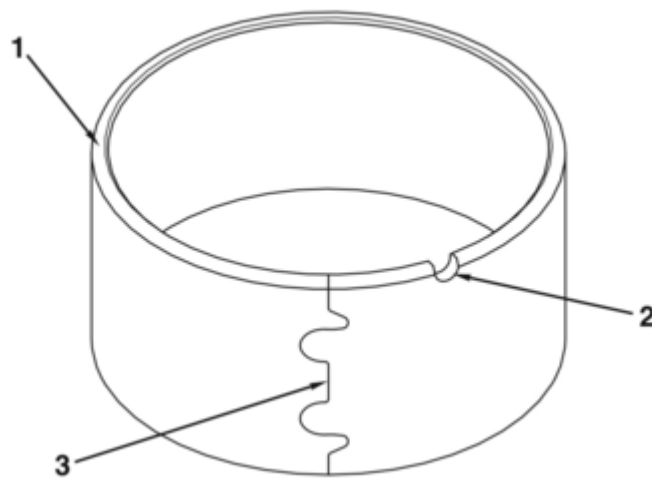


01r00074

Figure 3: Bushing with flange.

- 1. Flange
- 2. Butt joint.

©Cummins Inc

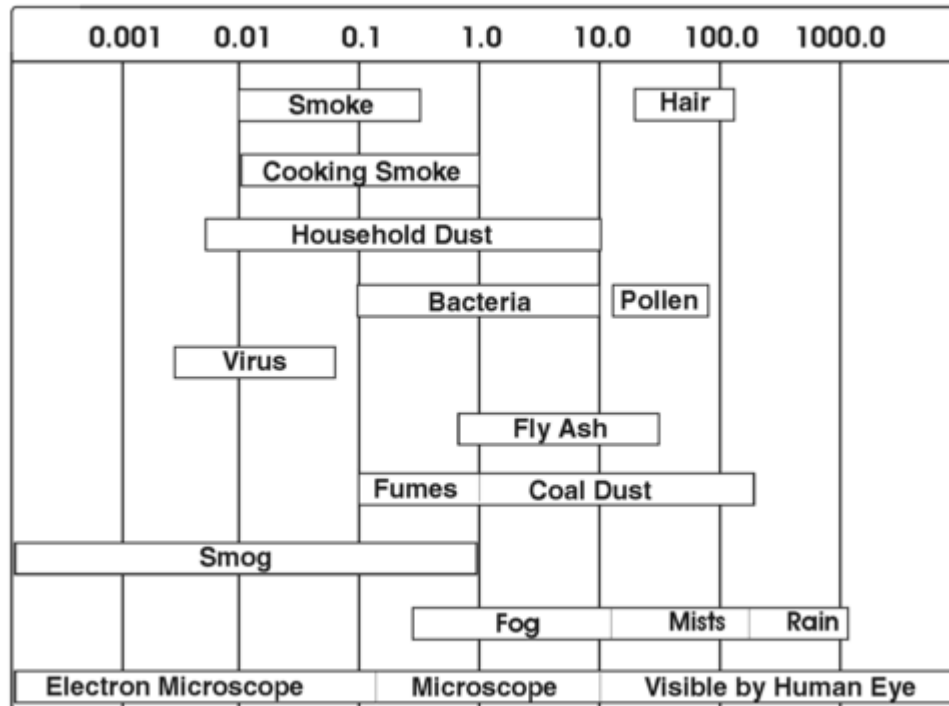


01r00075

Figure 4: Straight bushing.

- 1. End face
- 2. Locating notch
- 3. Clinch joint.

What is a Micron?



01r00076

Figure 5: Micron comparison.

The critical size is 5 to 30 microns. Contaminants in this range are larger than the operating clearances in an engine but too small for the oil filter to remove completely. These contaminants are responsible for an estimated 45 percent of the wear in an engine.

The correct oil filter will remove the majority of contaminant particles of 30 microns and larger that are suspended in the engine oil¹. Depending on the filter media, the filter also removes about half of the particles that are 10 microns and about 15 percent of the particles that are 5 microns. Some things to remember about oil analysis:

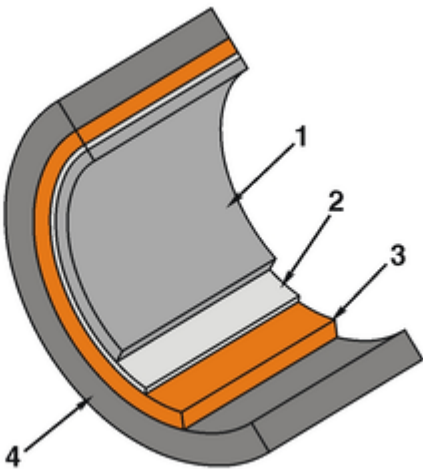
- Oil analysis looks for wear metals in solution.
- Oil analysis laboratories filter out particles above 3 to 8 microns.
- The particle has to be at least 10 microns to be visible without magnification.
- If the oil filter is full of chunks and the analysis history shows no problem, then bearing damage is probably progressive.
- Analysis for TBN (Total Base Number), TAN (Total Acid Number), and viscosity should be included to identify excessively long oil change periods.
- Do **not** forget to ask about the type of oil used/added between changes and the date of the last oil change. Did someone forget to add oil then start the engine?
- For best results from oil analysis, it is important to have a complete history of the analyses done on a specific engine serial number and to have an analysis of the new oil used by the customer.

Note(s):

1. Cummins® Service Bulletin Number 3810340, Cummins® Engine Oil and Oil Analysis Recommendations.

Construction

©Cummins Inc

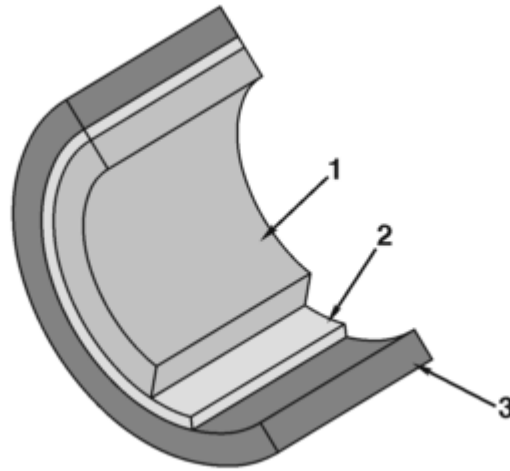


01r00077

Figure 6: Trimetallic bearing basic construction

Item	Description	Approximate Thickness in Microns (µm)
1	Overlay	15*
2	Nickel Barrier	2
3	Lining	300 to 500*
4	Backing	As required for backing.
* Dependent upon whether bearing is standard or over size.		

The bearings and bushings used in **all** Cummins® products are of trimetallic or bimetallic basic construction. Different types of bearings are used in different engine models and in some cases in the same engine model. Cummins Inc. continues to explore new bearings and materials in effort to remove the lead from bearings while maintaining high reliability and durability. Bearing types and materials are subject to change without notice. In addition, the suppliers of bearings and bushings may vary with Cummins Inc. and supplier manufacturing plant location.

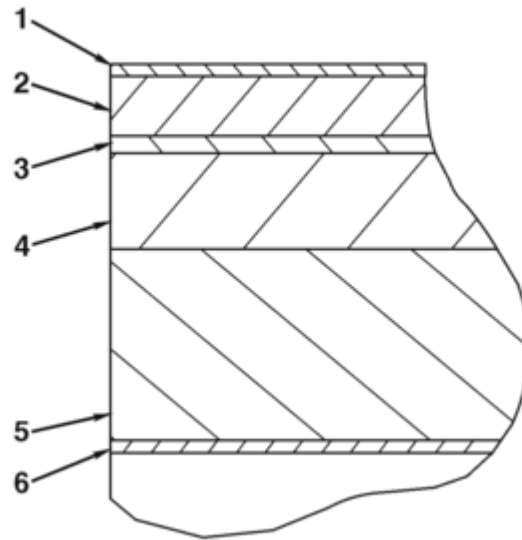


01r00078

Figure 7: Bimetallic bearing basic construction.

Item	Description	Approximate Thickness in Microns (μm)
1	Aluminum Alloy Lining	15*
2	Bonding Layer	2
3	Backing	As required for retention.
* Depends whether bearing is standard or oversize.		

Bearing and bushing construction usually begins with a strip of steel, called the backing. A softer lining material is applied with a uniform bond, which requires delicate control of feed rates and temperatures. The strip is cut into blanks and formed into the desired shape. A locating tang or notch is added, a parting line height cut, the oil holes are punched or drilled, and oil grooves are cut. The overlay is added to the trimetallic type bearing after **all** the features are completed. Some bearings then receive a tin-based flash plating for appearance and shipping and storage protection against corrosion.



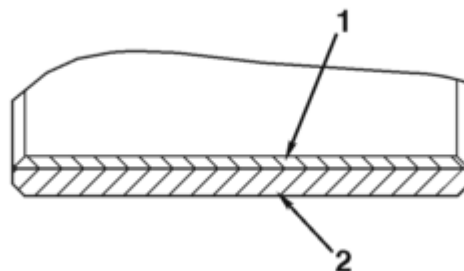
01/00079

Figure 8: Typical trimetal bearing cross-section.

Item	Description	Approximate Thickness in Microns (μm)	Function
1	Tin Flash	1.5	Corrosion protection during shipping and storage
2	Overlay (mostly lead with some tin and copper)	15	Conformability, embedability, corrosion resistance, and friction reduction. The overlay is added to the trimetallic bearing type for additional particle and marginal lubrication protection.
3	Nickel Barrier (nickel dam)	2	Prevents movement of material between layer 4 and layer 3 during bearing manufacture.

Item	Description	Approximate Thickness in Microns (µm)	Function
4	Lining (mostly copper with some lead and tin)	350*	Same as layer 2 but to a lesser degree. Adds to those functions plus some fatigue strength. The softer lining allows the bearing to conform to small surface and geometric variations in the bearing, shaft, and housing. Also, harmful particles can be embedded into the lining without extensive damage.
5	Back (low carbon steel)	As required for retention.	Provides shape. High fatigue strength. Provides the crush with the bore in the cylinder block, connecting rod, gear, etc.
6	Tin Flash	1.5	Some bearings will have tin flash on the inside diameter but not on the steel back.
* Depends whether bearing is standard or oversize.			

©Cummins Inc



01r00080

Figure 9: Typical bimetal bearing cross-section.

Item	Description	Approximate Thickness in Microns (µm)	Function
1	Lining (aluminum alloy)	400*	Conformability, embedability, corrosion resistance, and friction reduction.
5	Back (low carbon steel)	As required for retention*.	Provides shape. High fatigue strength. Provides the crush with the bore in the cylinder block, connecting rod, gear, etc.
* Depends whether bearing is standard or oversize.			

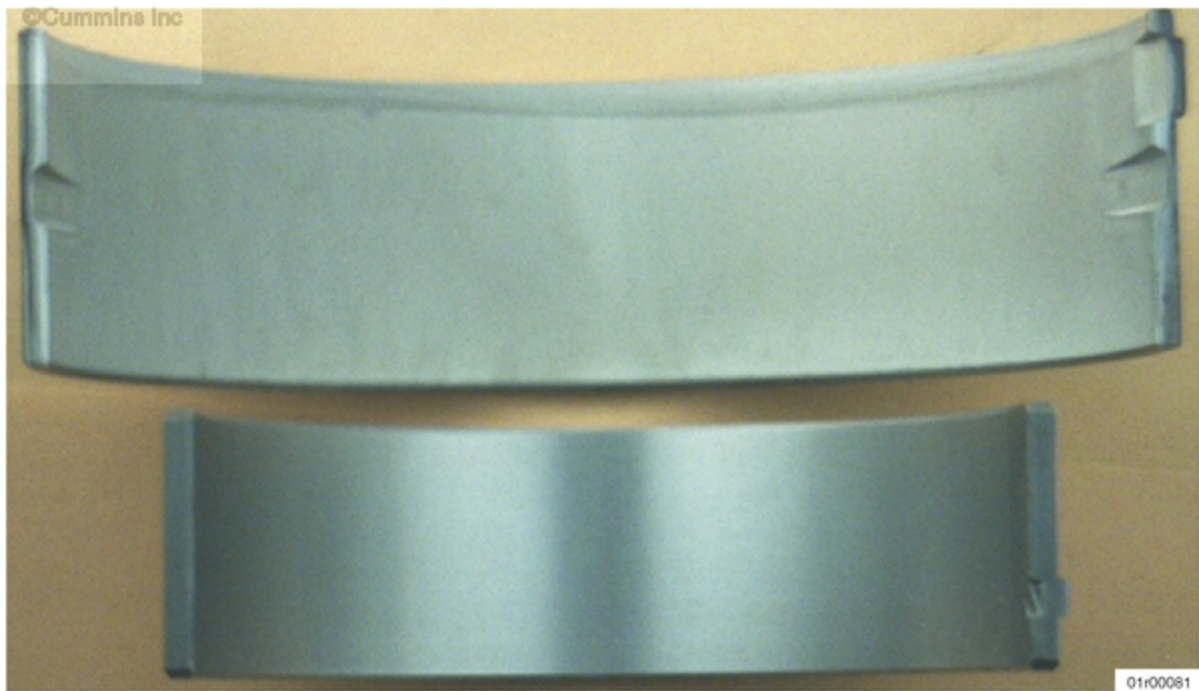


Figure 10: Trimetal (top) and bimetal (bottom) appearance.

Figure 10 shows a new trimetal (top) and new bimetal (bottom) lower main bearing from two different engines. Note the dull gray appearance of the tin flash plating on the trimetal bearing. The bimetal bearing has the appearance of bright aluminum.

Note : The line visible in the upper bearing is not a flaw in the bearing. It is the result of a wash during manufacture and does not affect the quality of the bearing.

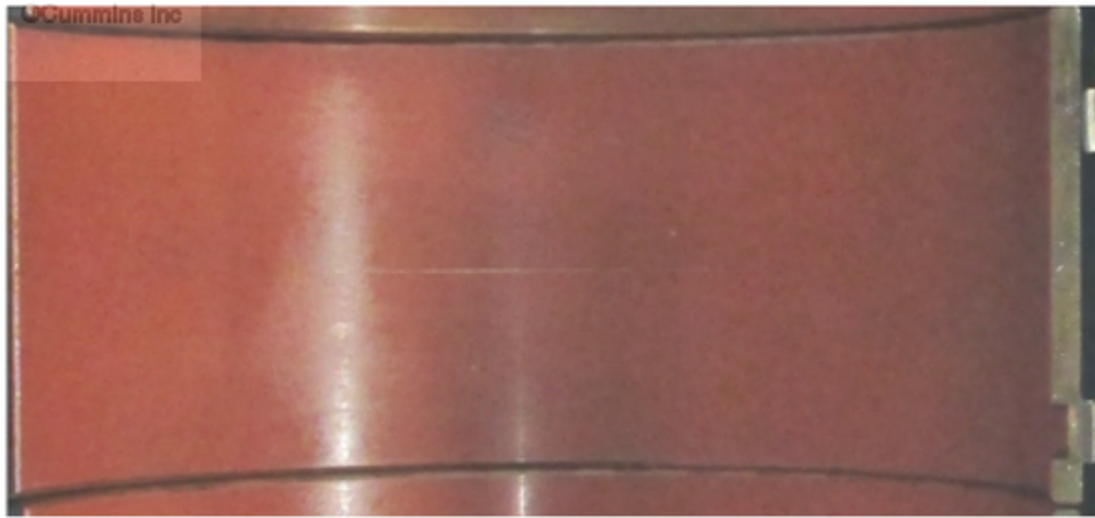


Figure 10A: Bimetal bearing with polymer overlay.

Figure 10A shows a bearing with a polymer overlay. The overlay contains a dry lubricant and other materials. The lining of this bearing is an alloy that is mostly copper. If any debris penetrates the polymer overlay, the copper will be exposed. This bearing meets the criteria for a “lead-free” bearing. Some engines may use a similar bearing that has an overlay of a different color.

Design

©Cummins Inc

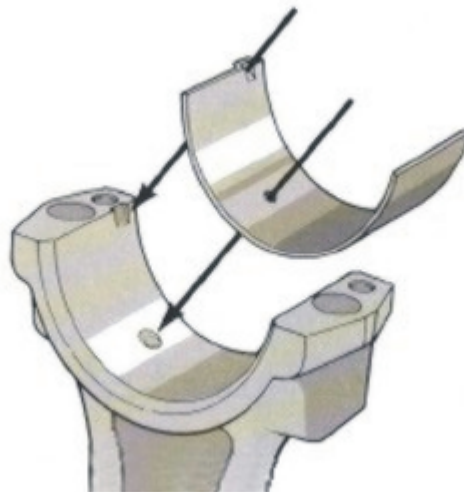


Figure 11: Locating tang and oil hole.

A bearing **must** protect and hold a shaft in place. It **must** withstand brutal punishment (pressure and temperature) yet absorb harmful contaminants. For the bearing to survive, it **must** be held firmly in place and in contact with the bearing bore. A locating tang, along with the free spread of the bearing shell, is the most common method of holding a shell type bearing in place during assembly. The tang should fit tightly in the slot in the mating part.



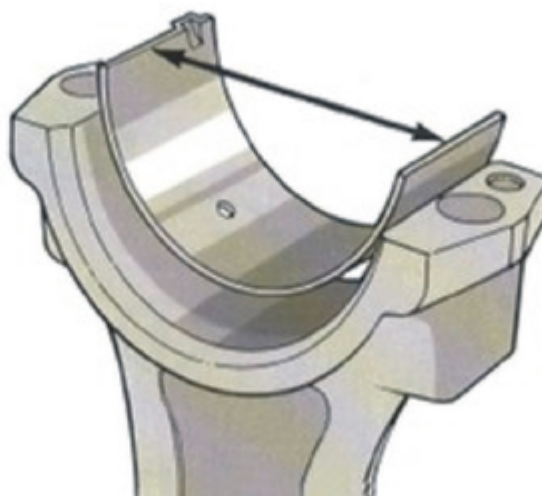
01r00084

Figure 11A: Locating tang on some bearings.

Some connecting rods of the fracture-split cap/rod joint construction utilize an additional tang to cause the alignment of the bearing parting line with the fracture joint.

Note : The locating tang and slot are **not** designed or intended to prevent the bearing from rotating in the bore after the main or rod bearing cap is installed. If the bearing(s) have rotated in the bore, it is an indication that the shaft and bearing have made contact without a sufficient layer of lubricating oil, there have been debris particles that are larger than the shaft to bearing clearance, or the clamp load on the cap and bore has been reduced.

Oil holes and grooves are **not** present in **all** bearings. If no oil holes or grooves are present, the bearing to shaft clearance distributes oil to the bearing and mating parts. When replacing bearings that contain oil holes, make sure the replacement bearing oil holes match both in the size and location of the oil holes in the in the old bearings and the oil hole in the mating part.



01r00085

Figure 12: Bearing shell free spread

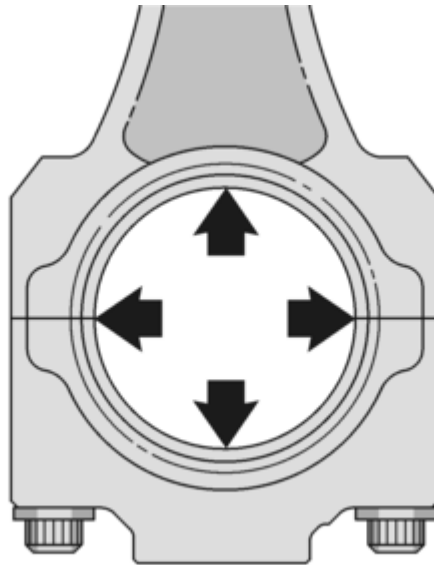
Main and connecting rod bearing shells are manufactured with "free spread". The distance between the parting faces is slightly larger than the diameter of the bore in the part. This free spread assures retention of the bearing during assembly and works with bearing overstand to provide a tight fit of the bearing against the bore when it is installed.



01r00086

Figure 13: Bearing shell overstand (crush height).

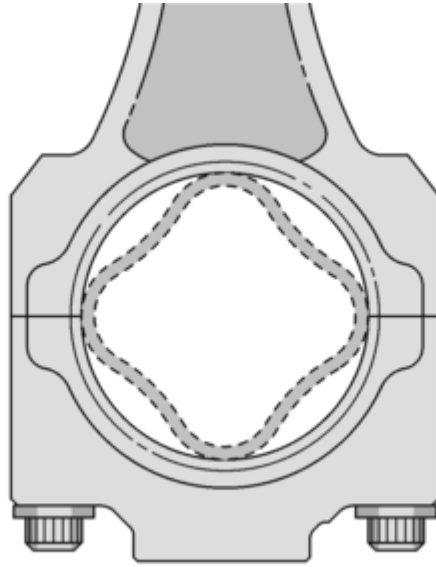
When a main or rod bearing shell is installed, the parting line will extend slightly above the parting face of the part. This is called overstand.



01r00087

Figure 14: Correct main and rod bearing crush.

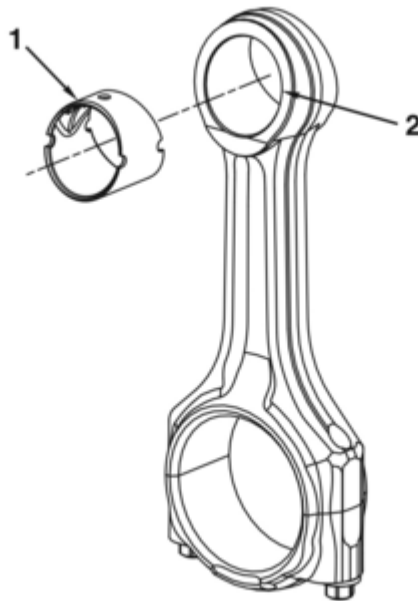
The overstand and free spread of the bearing combine to provide contact pressure (radial pressure) of the bearing shell against the housing. Crush height assures complete and tight contact with the housing bore when the capscrews are tightened to specification. This crush is what prevents the bearing shell from spinning in the bore unless there is heavy contact between the bearing and the shaft. Lack of crush allows the bearing shell to move in the bore causing excessive wear, fretting, and fatigue. Inefficient heat dissipation created by the lack of contact on the bearing back with the housing bore will increase bearing temperatures, increasing severity of operating conditions leading to accelerated bearing damage.



01R00088

Figure 15: Excessive rod and main bearing crush.

Excessive crush will bulge the wall of the bearing causing excessive wear on the surface at the parting edges and produce excessive heat concentration. This condition will cause the bearing to fail. Make sure the correct standard or oversize bearings are used in the bore.



01R00089

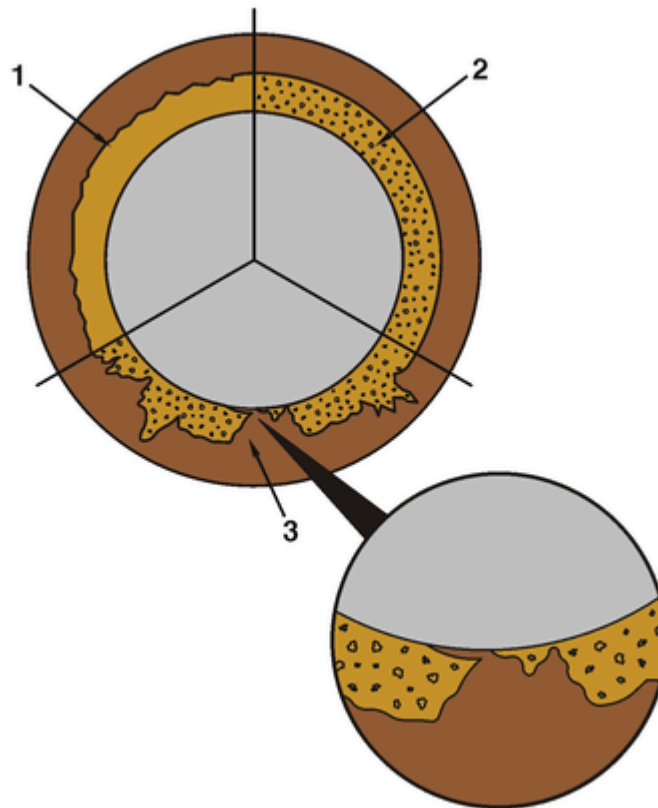
Figure 16: Bushing press fit.

All bushings in the engine are termed as “press-fit”. The example shown is a connecting rod bushing. The outside diameter of the bushing (1) is larger than the inside diameter of the connecting rod (2). The difference in diameter is what holds the bushing in place. It is critical that there be as much contact with the back of the bushing and the bore in the part as possible. This will prevent hot spots in the part which can cause the bushing to fail. The recommended practice for replacing bushings is to thermally shrink the bushing diameter by soaking in liquid nitrogen along with the installation arbor and increasing the mating part diameter by heating to 150 to 175°C [300 to 350°F]. This sort of assembly reduces the risk of removing material from the back of the bushing and has proven to provide for more consistent contact between the bushing and

the part. Make sure there are no burrs or sharp corners on the bore of the part. If there is material removed from the bushing during assembly, that bushing should be removed, the cause for the material removal repaired and a new bushing then installed. The bushings should be pressed into place using a hydraulic press or arbor press. Do **not** assemble the bushing with a tool and a hammer.

Bearing Surfaces

©Cummins Inc

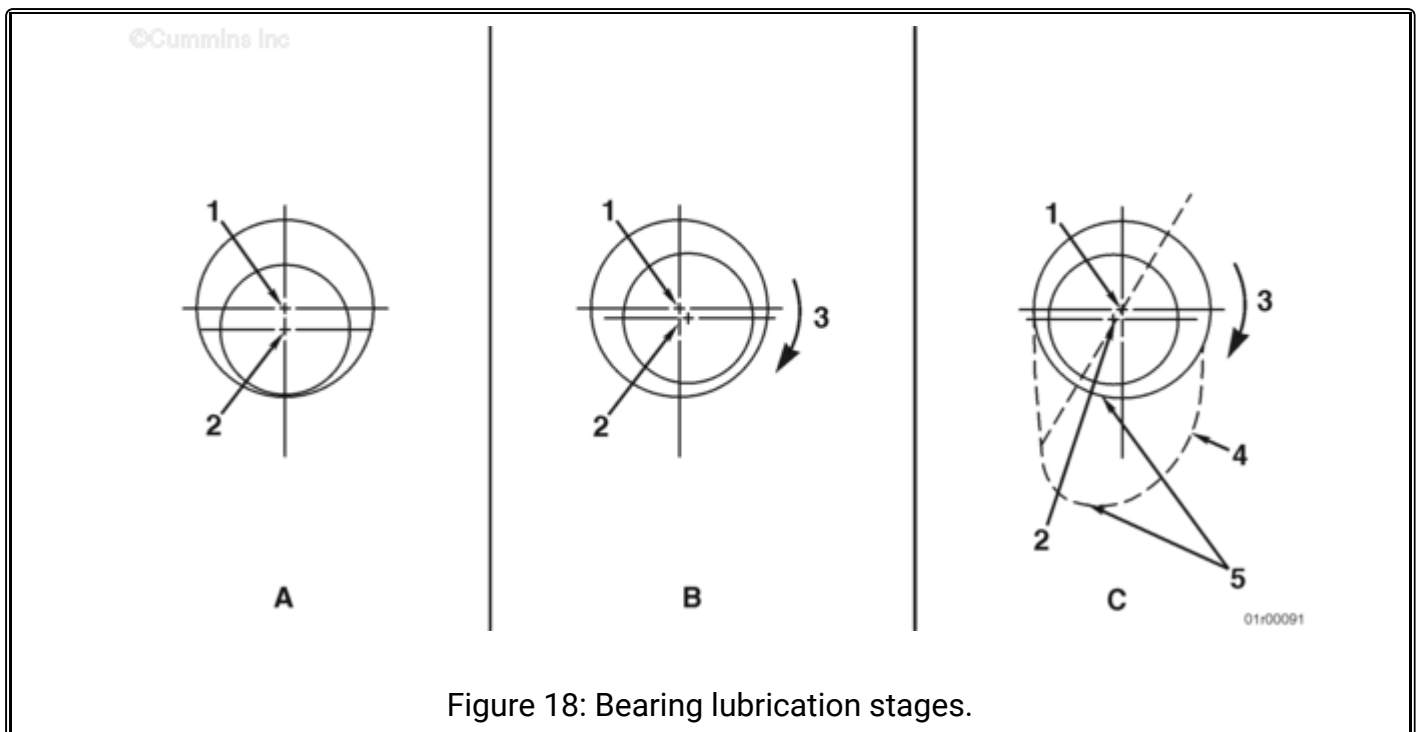


01r00090

Figure 17: Bearing surfaces.

No matter how smooth bearing and shaft surfaces appear, they are actually **not** perfectly smooth. Any surface contains peaks of material called asperities. Microscopic examination of the surface would show these asperities (1). Although very small, if the asperities on the shaft and bearing make contact, a process called micro-welding will pull material off the softer part. Friction caused by the asperities will quickly destroy the bearing and potentially the shaft. The insertion of a lubricant, usually engine oil, between the metal surfaces fills in the asperities and provides a layer of lubricant between the parts (2). However, if the surface of the bearing is scored or damaged with deep scratches or particles, the action of the lubricating oil will be impaired or defeated (3). The damage to the surfaces will change the oil distribution process and place excessive pressure on the edges of the damage. This will generate heat, which affects both the lubricant and the part, and the damage will progress. Clean and correct lubrication are the most important factors in bearing and bushing life.

Bearing Lubrication



1. Bearing centerline
2. Shaft centerline
3. Direction of shaft rotation
4. Oil film pressure depending upon speed and load
5. Maximum oil film pressure.

Frame A in Figure 18 shows the shaft at rest. The shaft is resting directly on a thin film of residual oil between the shaft and bearing. This is called “boundary lubrication”. There is no oil pressure. The oil layer and beneficial surface films are **only** a few microns thick, depending upon the temperature and how much time has passed since the shaft was turned.

Frame B in Figure 18 shows the motion of the shaft during engine start. As the shaft turns, it tries to climb up the wall of the bearing. Initially there is boundary lubrication, then oil begins to flow. This condition is called “mixed-film lubrication”. The oil film thickness is still **only** a few microns thick. The length of time this condition exists depends upon three factors; the length of time it takes for the lubricating oil system to supply oil to the bearing, the shaft speed, and the load. This condition is unavoidable during engine start up and is when a great percentage of the bearing wear takes place. The magnitude of the wear is dependent upon the delay time in delivering oil to the bearing, load, temperature, and oil viscosity. Bearing damage increases with oil delay time, increasing load, increasing temperature, and reduction in oil viscosity.

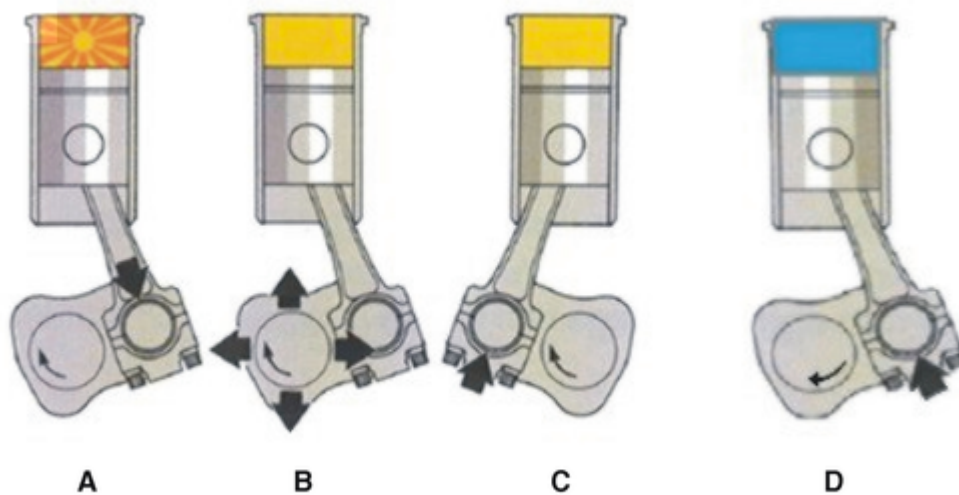
Frame C in Figure 18 shows the position of the shaft in the running position. The dotted line shows the oil film pressure distribution caused by the shaft rotation forcing oil between the shaft and bearing. This condition is called “hydrodynamic lubrication”. The oil flow into the bearing and shaft interface forms a wedge that prevents shaft-to-bearing contact. The thickness of the oil film at the most narrow point of the wedge depends upon shaft speed, load applied, oil temperature, and oil viscosity. The thickness may be less than one micron, depending upon the location of the

parts in the engine. The oil pressure at this point is calculated to be several thousand pounds per square inch (psi). Remember, however, that the oil wedge is constantly flowing and it is drawn into the wedge by the action of the shaft and the pressure in the system.

Bearing Load

Bearings, particularly the connecting rod and crankshaft bearings, are exposed to varying loads during engine operation. The load also varies depending upon the amount of work the engine is required to perform. The different loading of the connecting rod bearings is shown in Figure 19. The illustration is of a four-cycle engine.

©Cummins Inc



01r00092

Figure 19: Connecting rod bearing load.

See Figure 19:

- A. Combustion pushes down on the piston and connecting rod, applying a tremendous load on the upper bearing in the connecting rod.
- B. The connecting rod moving around the crankshaft will create centrifugal force that is considerable.
- C. When the downward inertia of the piston is reversed, force is again exerted on the top half of the rod bearing as during the exhaust stroke.
- D. During the intake stroke, the lower connecting rod bearing may be loaded. The amount of load depends upon the engine speed and boost pressure when the intake valve is open.

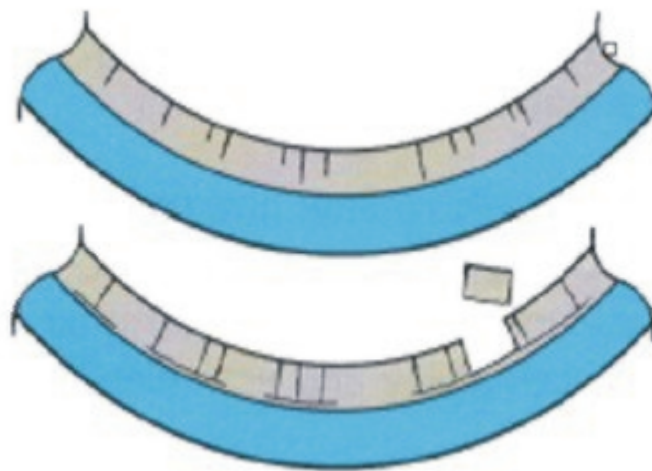
The lubricating oil film **must** be maintained through the complete cycle to protect the surfaces of the metal under these various stresses.

Metal and Material Properties

No single metal can provide **all** the properties demanded of a bearing. Therefore, bearings are a compromise combination of metals. Below is a list of the six bearing metal properties in their order of importance. The chart illustrates those qualities relative to the material used.

Property	Backing	Lining		Overlay
	100% Steel	Copper Lead (trimetal)	Aluminum Alloy (bimetal)	Overlay (lead tin) (trimetal)
Fatigue Strength	High	Medium to High	Low to Medium	Low
Conformability	Low	Medium	High	High
Embedability	Low	Medium	Medium to High	High
Corrosion Resistance	High	Low to Medium	High	High
Friction Reduction	Low	Medium	Medium	High
Heat Resistance	High	Medium	Medium	Low

©Cummins Inc



01r00093

Figure 20: Bearing lining fatigue.

Fatigue Strength

Bearing fatigue is the gradual deterioration of the bearing lining metal caused by loading. The first signs of fatigue are very small cracks on the surface of the bearing lining. Cracking of the overlay of trimetal bearings is normal. The lining fatigue cracking can move through the lining to the bond line between the lining and back then turn and propagate along the bond line. The cracks eventually connect. The hydraulic pumping action of the oil will cause metal to flake away, exposing the steel back. This is sometimes mistaken for a bad bond but the very rigid controls used during bearing manufacture make a bad bond improbable. The steel back will remain because it provides optimum fatigue strength.

Conformability

No crankshaft journal, crankshaft pin, shaft, or gear is perfectly round or straight nor are the bores for the bearings or bushings. To overcome these slight imperfections, the bearing **must** be able to mold itself to deform with these imperfections. The softer materials of the lining and overlay are necessary for conformability. Once a pair of crankshaft bearings has been used in one of the bores in a cylinder block with a specific crankshaft, it **must only** be reused or reinstalled in that bore with that crankshaft. Once a pair of connecting rod bearings has been used in a connecting rod, it **must only** be reused or reinstalled in that same connecting rod on that same crankshaft pin. If the bearings are **not** marked for location in the engine when they are removed, they **must not** be reused. Some bearing materials are **not** suitable for reuse and are labeled "Do Not Reuse" on the bearing back.

Embedability

No matter how careful you are and how excellent a filtering system is provided, some debris will be present. Particles or debris can score the shaft or bearing. If the bearing surface can absorb some small quantity of these particles, further damage can be prevented. Softer lining materials permit conformability and embedability.

Corrosion Resistance

The by-products of combustion, which include water and other chemicals, along with the degradation of the oil itself, combine to form harmful acids and compounds which can attack the bearing materials. The materials used each have a specific corrosion resistance but can be attacked if the engine operating conditions produce more of the acids and compounds than normal and/or the oil change interval is too long.

Friction Reduction

Since it is impossible to prevent **all** contact between a bearing and a shaft, it is desirable to have two materials which can operate in contact. A material which can resist seizure under such conditions is desirable to help prevent failures when lubrication is marginal.

Heat Transfer

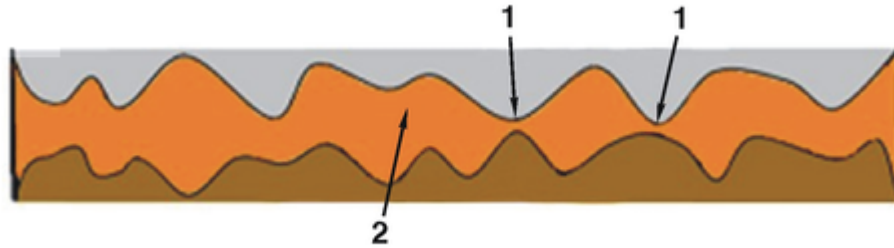
Heat is created by the friction which exists in movement. While much of the heat is removed by the oil flow through the bearing, some of it is conducted away through the housing and the shaft. A good conductive material is desired along with good contact with the bearing bore.

What Causes Bearings to Fail

All bearing failures, regardless of bearing type, are the result of excessive shaft to bearing contact. Excessive shaft to bearing contact creates heat. Excessive heat will damage the bearing or the shaft (or the race if a roller bearing). The damage is cumulative. That is, once the bearing and/or shaft surface is damaged, further contact will likely increase the damage. This cumulative damage can be milliseconds to many hours later and will depend upon the applied load and oil film thickness.

No shaft or bearing is perfectly smooth. The surfaces have microscopic peaks and valleys. The peaks are called "asperities". The major function of the lubricating oil (or fluid in the case of fuel pumps), is to form a film or barrier that will prevent contact between the asperities.

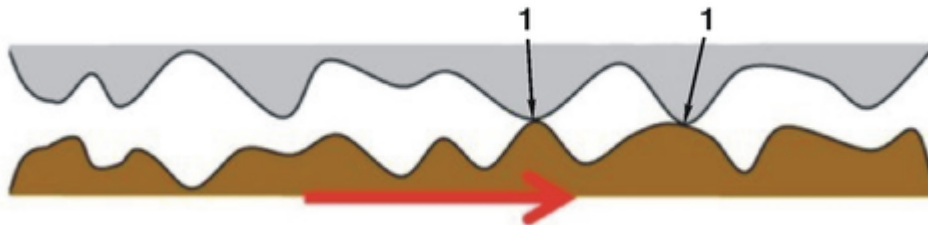
©Cummins Inc



01r00094

Figure 21: Bearing asperities (1) with oil film (2).

©Cummins Inc

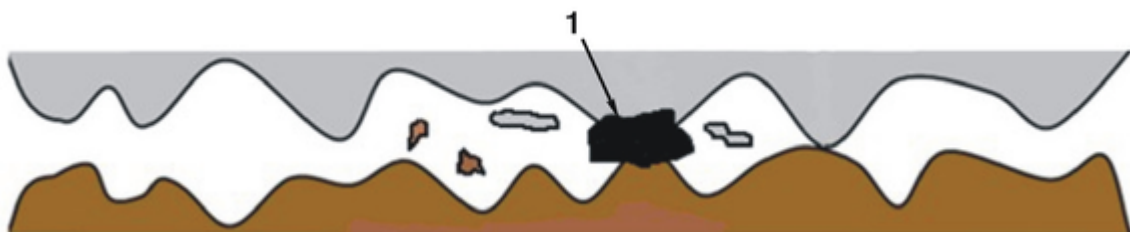


01r00095

Figure 22: Asperity contact (1) (inadequate oil film).

If the shaft and bearing make contact, it is due to the lack of effective lubrication. It indicates that the layer of oil is **not** thick enough, for whatever reason, to prevent the high points, the asperities (1), in the shaft and bearing to make contact. At low contact pressure, the interaction causes friction and wear. As the contact pressure increases, micro-welding occurs. The shaft will continue to turn, pulling the welded material from the weakest/softest part, which is nearly always the bearing.

©Cummins Inc



01r00096

Figure 23: Dirt particle larger than oil film.

A particle in the oil (1), that is larger than the oil film thickness becomes an asperity. Whether the shaft and/or bearing are damaged depends upon the hardness of the particle relative to the hardness of the other parts. If the particle is hard enough and large enough to damage the

bearing and/or the shaft, the oil film will be reduced, possibly allowing more contact of the asperities.

The loss of the oil film can be the result of several different conditions as listed in the following table.

Oil Film Loss	
Condition	Potential Cause
"Global" or general lack of lubrication	• No or low oil level in oil pan • Oil pump stopped working • Broken oil suction tube • Operating angle greater than design • Large internal oil pressure leak • Large external oil pressure leak • Oil pressure regulator malfunction • Excessive oil degradation • Excessive oil delay due to incorrect oil for ambient condition.
Oil too thin (low viscosity)	• Incorrect oil for operating environment • Oil too hot • Oil contaminated with fuel, coolant, or transmission oil.
Oil too thick (high viscosity)	• Incorrect oil for operating environment • Oil contaminated with soot • Oil degradation (extended drain interval) • Oil contaminated with gear lubricant.
Fast Start	Engine speed (rpm) is increased too quickly immediately after starting
Cold Start	Engine speed (rpm) increased before cold oil can reach the bearings • Prelube not done prior to starting large engine • Failure to fill new oil filters with clean oil.

Oil Film Loss	
Condition	Potential Cause
Bearing or shaft surface damaged	• Foreign particles larger than the oil clearance between the bearing and shaft ◦ Dirt in block oil drillings ◦ Dirt in shaft oil drillings ◦ Oil filter plugged causing oil filter bypass valve to open ◦ Oil filter bypass valve stuck open (oil filter bypass valve opens every time the engine starts, except QSK95) ◦ Dirt induced during previous repair.
Incorrect Assembly	• Insufficient or incorrect assembly lubricant applied to bearings or shaft • Foreign particles between the bore and the bearing • Bushing backing material removed during installation • Locating tang not in the correct location • Wrong bearing part number • Inadequate clamp load on fastener • Dirty parts • Oil filter not primed • Engine not pre-lubed after a repair that removes oil from the system • Bore for the bearing is too large or too small.

Diagnostic Procedures

Experience has proven that many failures that are diagnosed as bearing failures are actually progressive damage from other failures in the engine. This is particularly true of failures that create debris or previous repairs that allow dirt and debris to enter the lubricating oil system. For these reasons it is important to look at and consider the entire engine history, **not** just the bearings. In many cases analysis of the bearings will lead to looking for other failures that may have nothing to do with the bearings.

Note : Do not destroy evidence by focusing only on the bearings.

Note : Go slowly and observe all conditions of the engine.

Note : Experience has proven that more will be learned from the bearings that did not fail.

Note : Experience has also shown that the analysis will seldom be able to state absolutely, positively the assignable cause of the failure. Most often, the analysis will offer a likely assignable cause after review of **all** the parts and using the process of elimination based on the condition of the parts.

While the customer complaint that might indicate a problem with a bearing or bushing can vary greatly, the most likely complaints are:

- Low oil pressure
- Unusual or loud noise from different areas of the engine
- Engine stopped and would **not** start (cranks too slow)
- Engine stopped and is locked
- Internal parts have escaped through holes in the cylinder block and/or oil pan.

When these complaints occur, the following process is recommended:

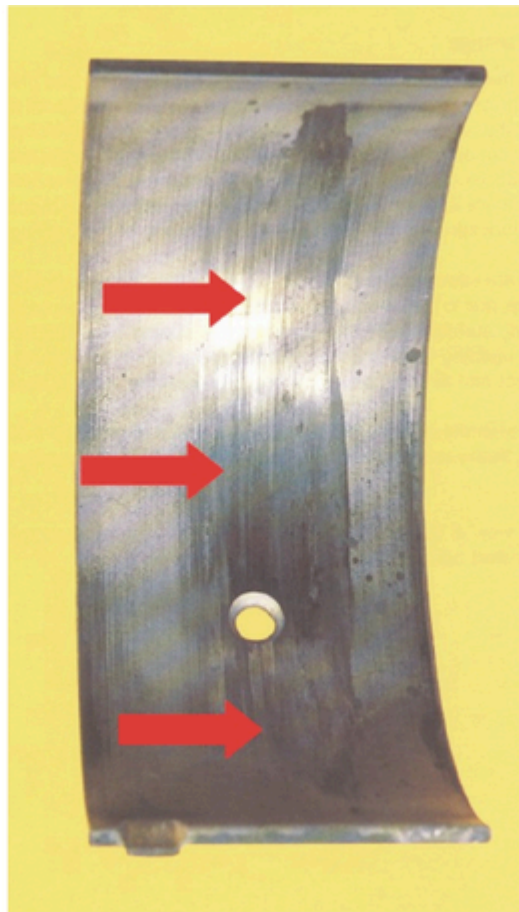
- Find and record the ESN (Engine Serial Number)
- Ask the operator exactly what the engine was doing when the problem occurred
- Check and record the oil level using the dipstick. Look for any presence of liquid other than engine oil on the dipstick.
- Disconnect the power to the starting motor
- Attempt to rotate the engine manually and note the amount of physical effort required
 - The effort to rotate the engine is “about right” in the experience of the technician
 - The effort to rotate the engine is “too much” in the experience of the technician
 - The engine can **not** be rotated manually.
- Measure the crankshaft end clearance
- Remove and examine the oil filter.
 - Obtain an oil sample from the filter
 - In this examination we are looking for large quantities of debris
 - Use a magnet to determine if the debris is ferrous or non-ferrous
 - Examine the plates of the bypass filter section of combination filters in the same way as the full-flow section pleats
 - On engines with multiple filters, start with the filter closest to the oil inlet, but look at **all** the filters.
- Obtain an oil sample from the oil pan
- If the complaint is low oil pressure, the engine will rotate manually, and the oil filter appears to be acceptable, start the engine and troubleshoot for low oil pressure
- With any of the other complaints:
 - Gain access to the crankcase
 - Look at the main bearing and connecting rod caps for signs of excessive heat
 - Look for signs of heat from regions near cam bushing and gear bushings
 - Check **all** the rods for the presence of side clearance.
- The next steps will depend on the conditions found thus far.
- **Note : Avoid wiping the lining side of the bearing with anything. Wipe and mark the back of the bearing shell.**
- As the bearings are removed, mark them for location by cylinder and/or main cap and position (upper or lower). This is particularly important if nothing wrong is found and the parts can be reused per the guidelines later in this document.

- Clean the bearings with solvent. Use **only** your fingers to gently remove the oil from the overlay/lining.
- Lay **all** the bearings out as they are in the engine
- Record **all** the part numbers, vendor code and date codes for each bearing shell
- Examine the bearings using the examples found in this document and make note of the condition of each
- Examine the shaft, bearing bores, cap faces, etc., for condition
- Pay very close attention to find “witness marks” or disturbed machining marks on the parts associated with forming the bore for the bearing or bushing.

Bearing Damage Patterns

Wiping:

©Cummins Inc



01r00097

Figure 24: Example of heavy wiping on a trimetal bearing shell.

Note : The bearing in Figure 24 does not have a nickel layer between the overlay and the lining.

The surface of the trimetal rod bearing in Figure 24 is shows wiping damage. The lubricating oil pump on the engine had failed. Heavier damage was prevented by a low oil pressure shutdown device.

Wiping is the smearing or heavy rubbing of the bearing surface due to metal to metal contact with the journal surface. Wiping is the term usually applied to the initial stages of damage due to momentary lack of lubrication. Figure 24 shows what can be termed as heavy wiping.

Note that the wiping shown is **only** of the tin flash and lead based overlay. If the wiping had been heavier, then there would have been exposure of the copper-colored lining.

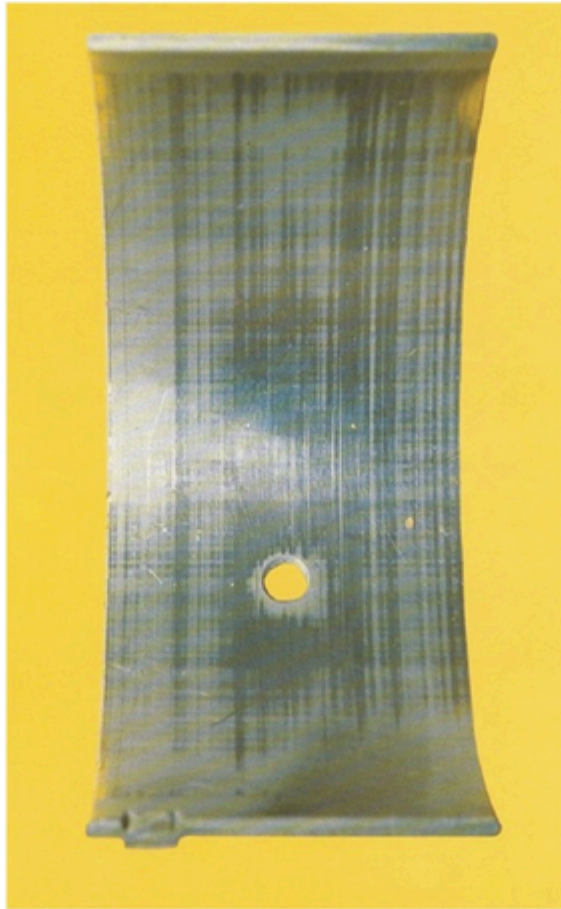
Some lighter wiping than shown in Figure 24 can be expected during disassembly of the rod and main bearings caused during metal-to-metal contact when the crankshaft is rotated to gain access to the connecting rod fasteners on each crankshaft throw.

During assembly, this sort of wiping is prevented by proper lubricating of the bearings per the appropriate service manual.

Note : Never apply lubricant to the back of rod or main bearings or camshaft bushings. Lubricating the back will interfere with the retention of the bearing in the housing.

Wiping damage occurs when there is a momentary loss of lubricant flow. Loss of flow for a longer duration or repeated momentary flow loss results in other patterns. Small amounts of air leakage into the suction side of the lubricating oil pump can occur when operating with low oil levels or a crack in the suction tube. A major cause of momentary lack of lubrication is operating the engine at high crankshaft speeds immediately after starting. It can take several seconds for sufficient oil to flow through the engine after starting depending on engine temperature at shutdown and how long the engine has been shut down. If the engine is accelerated to high crankshaft speeds within a few seconds after starting and before sufficient oil flow is established, the chances are that some bearings will **not** receive adequate lubrication to prevent wiping.

Note : Experience has shown that wiping of this sort will occur first on upper rod bearing shells, particularly if the engine has repeated fast starts.



01r00098

Figure 25: Example of moderate wiping on a trimetal connecting rod bearing shell.

Figure 25 exhibits what is called moderate wiping on an upper connecting rod bearing shell, caused by a momentary lack of lubrication. This bearing does **not** have a nickel dam.

The tin flash and overlay are somewhat smeared but there is no exposure of the copper color of the bronze overlay.

Figure 26 shows a photograph of **all** the bearings from the same engine.

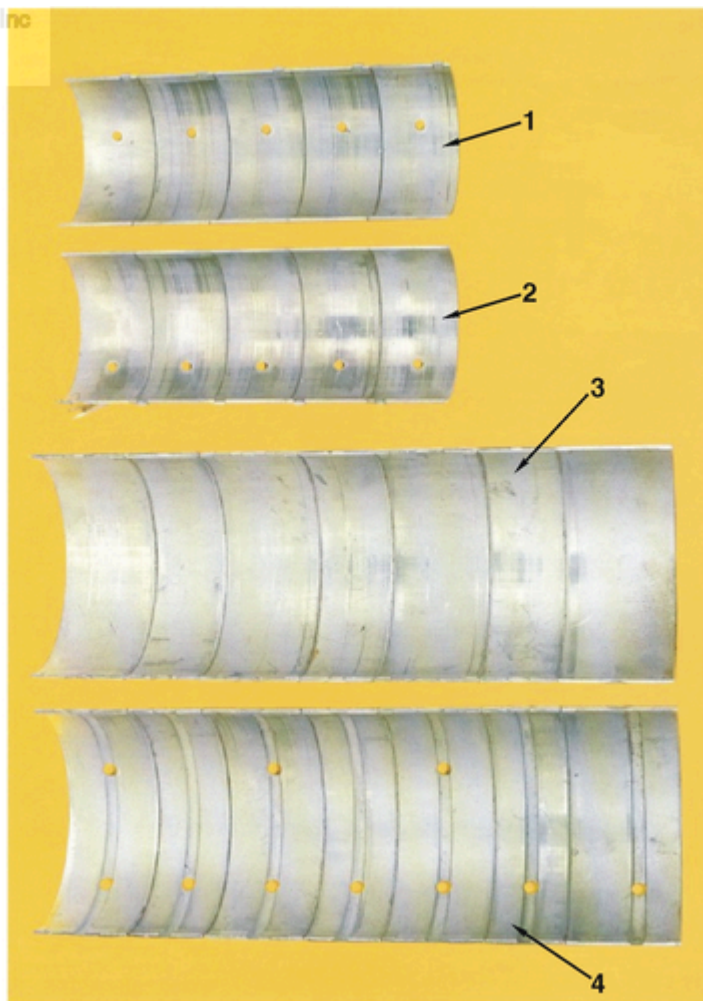


Figure 26: Example of moderate wiping on trimetal connecting rod and main bearings.

Figure 26 shows **all** the bearings returned from the same engine as Figure 25. The number 4 rod bearing had spun and was totally destroyed. Unfortunately, the number 4 connecting rod was **not** returned, so additional important information regarding this failure was **not** available. In addition, the bearings were **not** marked for their location.

1. Lower connecting rod bearings
2. Upper connecting rod bearings
3. Lower crankshaft bearings
4. Upper crankshaft bearings.

It is important to note that none of the bearings in the photograph are failed. The damage is largely cosmetic.

It is important to note that the wiping is more severe on the upper connecting rod bearings than the lower. This is due to the increased load created by cylinder pressure during the power stroke of the engine operating cycle.

The lower connecting rod bearings in this photograph exhibit more wiping than normal.

Note that there is less wiping on the lower crankshaft bearings than on the connecting rod bearings. The lower crankshaft bearings carry the heaviest load of the crankshaft. The lower crankshaft bearings exhibiting less wiping are as expected for a momentary loss of oil pressure, particularly if the engine operating at or near rated speed. The rod bearings are the first to show

lack of lubrication during a lack of lubrication event because they are at the end of the lubricating oil routed from the main oil rifle, through the cylinder block, the oil groove in the upper main bearing, and the oil drillings in the crankshaft.



Figure 27: Minor wiping of trimetal connecting rod bearings.

Figure 27 shows the upper and lower connecting rod bearing shells from a QSX15 engine that suffered a failed lubricating oil pump at two hours. The engine was shut down before any bearings were damaged. The bearings in the photograph could have been reused if they had been marked for location and installed in the same connecting rod on the same throw of the same crankshaft. This is a trimetal bearing with a nickel layer but the nickel layer is **not** visible.

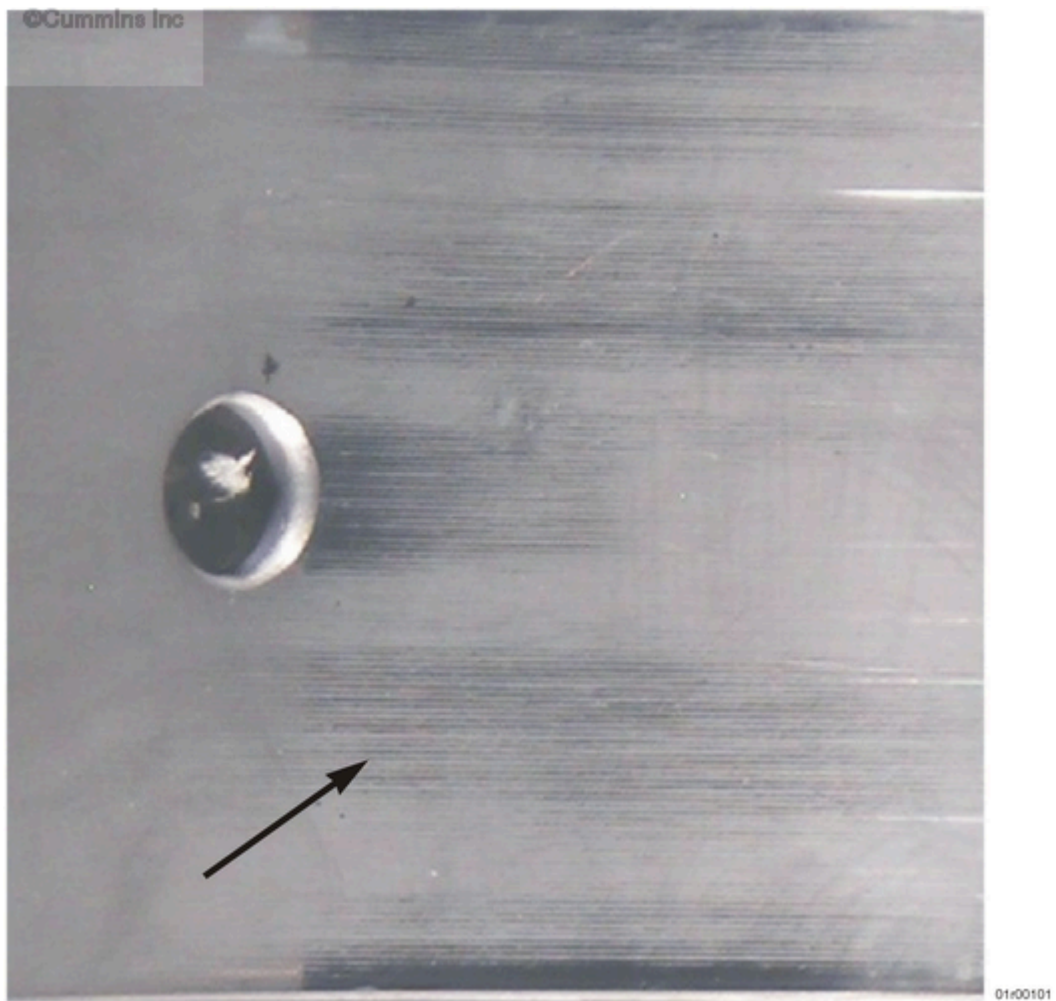
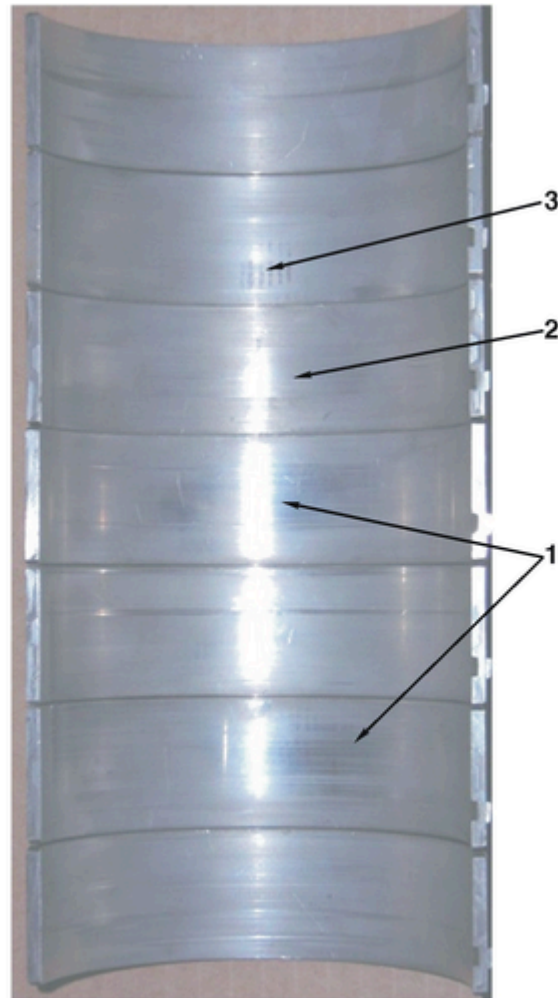


Figure 28: Close-up of minor wiping on trimetal connecting rod bearings.

Figure 28 is a close up of the third upper connecting rod bearing from the top in Figure 27. Note the straight lines in the areas of wiping. The straight lines are the result of the manufacturing process and are **not** damage from debris. In figures 27 and 28 the tin flash has been smeared by minor contact between the shaft and bearing. The overlay has barely been touched. This bearing has a nickel layer but the nickel layer is **not** visible.



01r00102

Figure 29: Wiping of bimetal lower main bearing shells.

Wiping of the lining of bimetal bearing shells is **not** as noticeable as on trimetal bearings.

These bearings are from an engine that had suffered a broken main bearing cap.

Item number 1 in Figure 29 exhibits moderate wiping and is **not** a failure.

Item number 2 in Figure 29 exhibits light wiping and is **not** a failure.

Item number 3 in Figure 29 is the bearing that was in the broken main cap. The vertical lines in that bearing shell are the result of handling and **not** the broken cap. The connecting rod bearing shells from this engine were **not** returned.

Loss of Overlay:

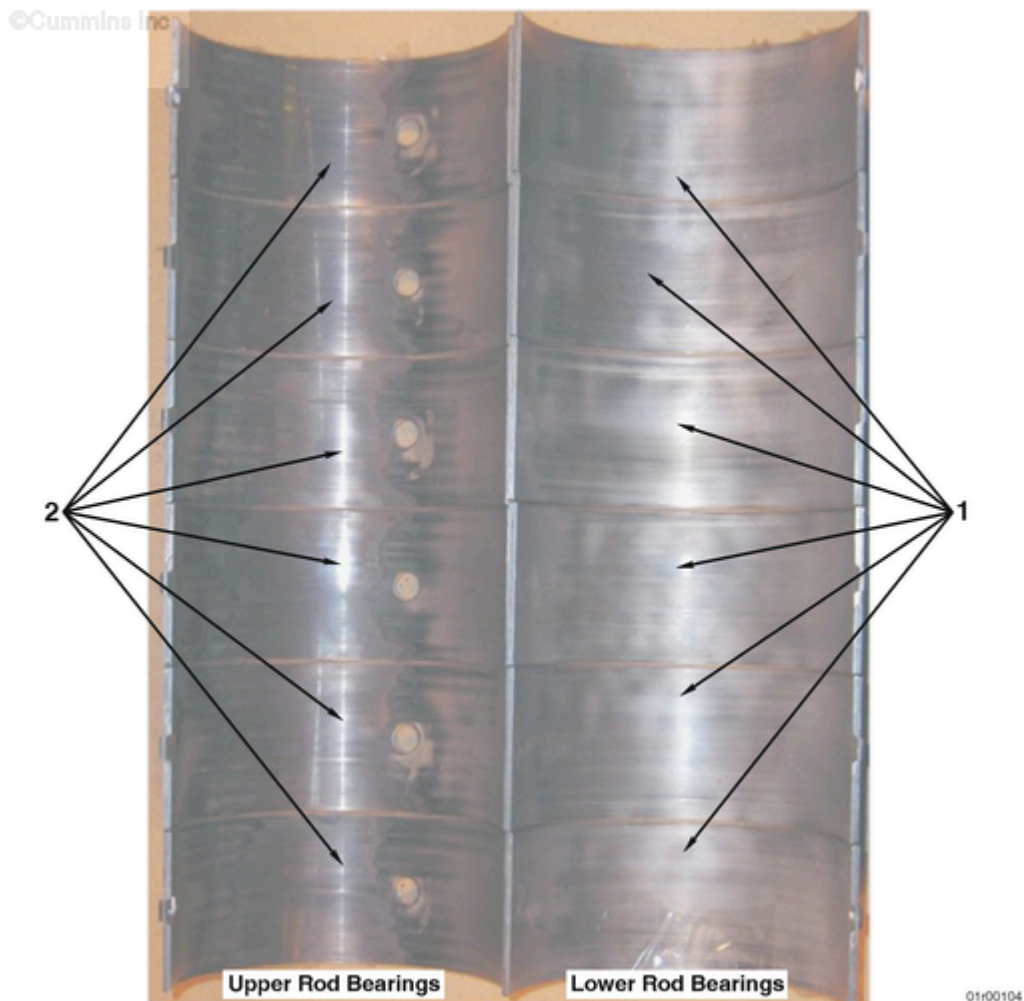


Figure 31: Overlay loss trimetal rod bearings with nickel layer.

Figure 31 shows the loss of overlay material on trimetal bearings that do have a nickel layer. These bearings are from an engine with 1,882,932 kilometers [1,170,000 miles] and were replaced at scheduled overhaul.

The dull gray (1) areas indicate where the overlay material is still present. Note that almost **all** the overlay is still present on the lower bearing shells but **only** at the mostly unloaded areas on the upper shells. The brighter areas (2) indicate where the overlay was worn away. Note there is no visible indication of exposure of the copper/lead lining material. The overlay has been worn enough to expose the nickel layer but no further.

Some degree of bearing overlay wear is expected. The bearings shown in Figure 31 do **not** indicate a problem unless the engine has relatively few miles or hours. If this sort of wear occurs at low miles or hours then operating and maintenance practices **must** be examined. Each of the following conditions will have a significant impact on the loss of overlay material:

- Operating the engine with low oil level
- Wide open throttle starting
- Long oil change intervals
- Overspeeding or over fueling.

Note : Air inlet restriction and exhaust restriction reduces the amount of air available and is, in effect, overfueling.

- Clogged oil filters

- Cooling system issues that allow the oil to be too hot.
- Excessive fuel in the engine oil.

Note : The bearings in Figure 31 have **not** failed.

Note : Bearings of bimetal construction do **not** have an overlay layer.

©Cummins Inc

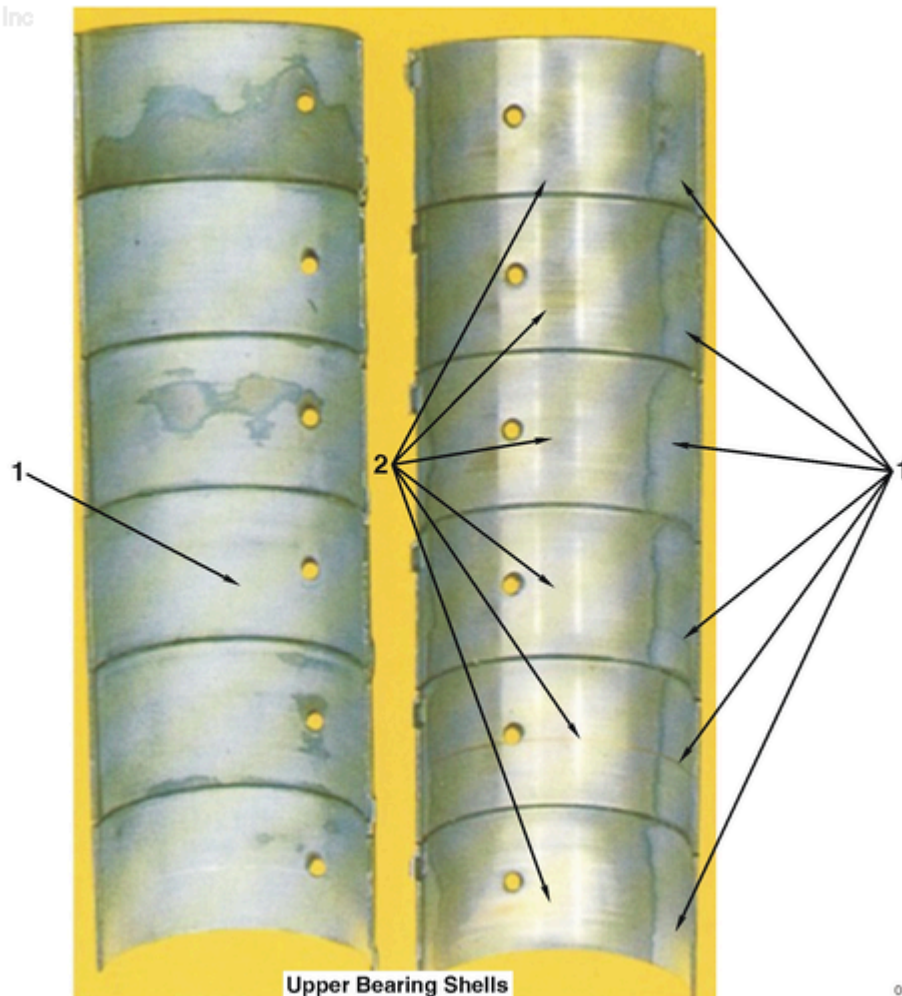


Figure 30: Loss of overlay material trimetal rod bearings.

Figure 30 shows the loss of overlay material on connecting rod bearings that do **not** have a nickel layer. The dull gray (1) areas indicate where the overlay material is still present. Note that almost **all** the overlay is still present on the lower bearing shells but **only** at the mostly unloaded areas on the upper shells.

The brighter areas (2) indicate where the overlay has been worn away enough to reveal some of the copper-colored lining material.

Loss of bearing overlay material occurs when mild momentary lack of lubricant flow is encountered on a repeated basis. In this situation, the metal-to-metal contact and the lack of lubrication causing it are **not** severe enough nor of a long enough duration at each occurrence to raise the bearing temperature excessively. This results in an absence of discoloration or other signs of overheating. However, each time metal-to-metal contact occurs, a small amount of the bearing surface material is worn away so the lubricating clearances are enlarged, causing a loss of lubricant retention. The bearings then become more vulnerable to damage.

Figure 30 shows a substantial loss of bearing overlay. The loss of overlay occurs first and more quickly on the rod bearings, particularly the upper rod bearings, because they are at one of the ends of the oil system.

Some degree of bearing overlay wear is expected. The bearings shown in Figure 30 do **not** indicate a problem unless the engine has relatively few miles or hours. If this sort of wear occurs at low miles or hours then operating and maintenance practices **must** be examined. Each of the following conditions will have a significant impact on the loss of overlay material:

- Operating the engine with low oil level
- Wide open throttle starting
- Long oil change intervals
- Overspeeding or over fueling

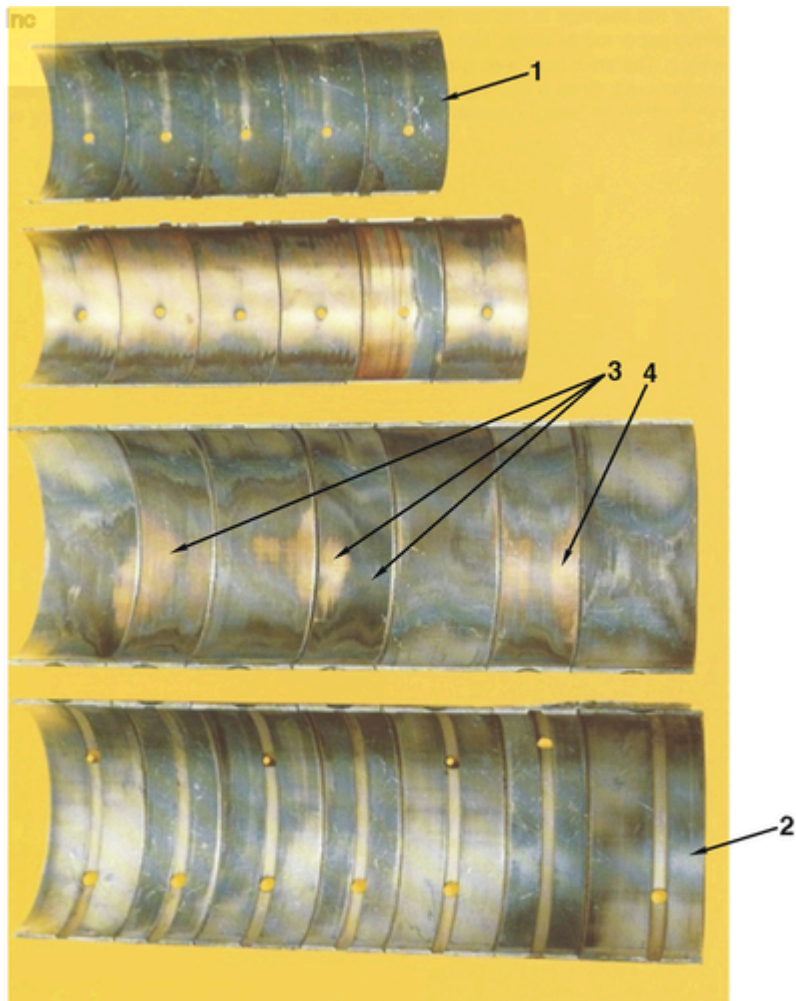
Note : Air inlet restriction and exhaust restriction reduces the amount of air available and is, in effect, overfueling.

- Clogged oil filters
- Cooling system issues that allow the oil to be too hot.
- Excessive fuel in the engine oil.

Note : The bearings in Figure 30 have not failed.

Note : Bearings of bimetal construction do not have an overlay layer.

Deep Wear:



01R00106

Figure 32: Deep wear on tri-metal bearing shells.

Repeated mild wiping conditions result in deep wear into the bearing lining exposing large areas of copper colored material in trimetal bearings. Figure 32 is a good example of this sort of wear. Many incidents of metal-to-metal contact are necessary to cause this sort of wear. Logically, metal-to-metal contact can **only** occur in the absence of oil film or when there is a shortage of lubrication. These particular bearings do **not** have a nickel layer between the overlay and the lining.

However, the absence of discoloration or other signs of overheating indicate that the wiping conditions that existed were of very short duration, such as can occur:

- In a fast start situation (operating the engine at a fairly high speed immediately after starting)
- Intermittent lowering of oil pressure that can occur when operating with low oil level or clogged oil filters
- During over-speeding and/or over-fueling, causing excessively high oil temperatures. The elevated oil temperatures, along with the additional load, reduces bearing oil film thickness to the point that metal-to-metal contact can occur.
- With higher oil temperatures and thinner oil film thickness, fine dirt wear is more pronounced
- Excessive fuel in the engine oil.

One of the upper rod bearings was **not** returned.

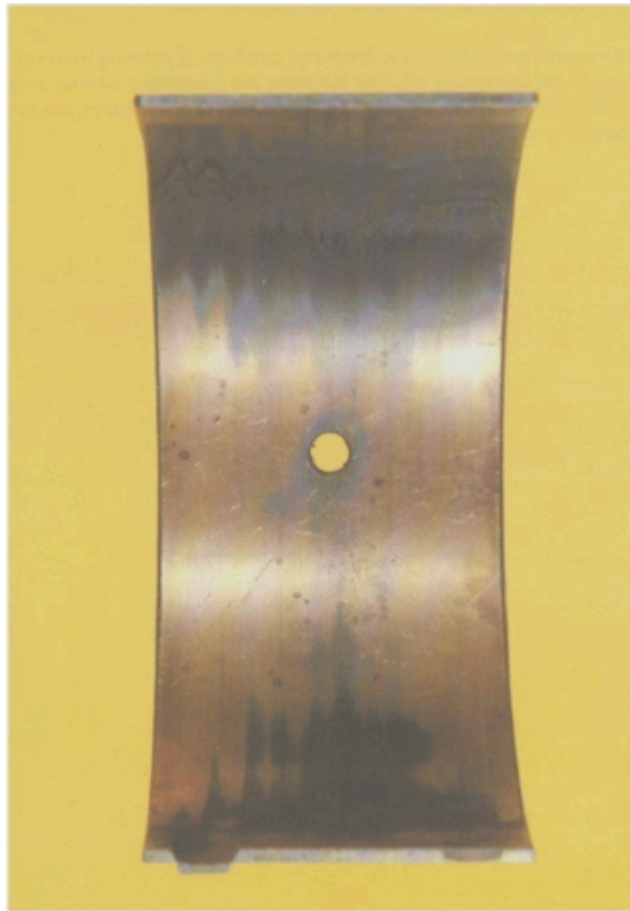
The dark staining seen on the lower rod bearings (1) and upper main bearings (2) is from varnish and insolubles in the lubricating oil. This sort of staining is indicative of long oil change intervals.

Exposed copper in good condition and free of fatigue cracks is **not** a failure. Bearings in this condition will continue to operate satisfactorily under normal operating conditions and with correct maintenance.

The exposed copper pattern on some of the lower main bearings (3) are due to minor geometry issues with the bore. Wear of this sort is **not** uncommon and is **not** a failure.

The exposed copper pattern on the lower main bearings (4) indicates a slight geometry issue with the crankshaft. Again, this pattern is **not** indicative of an impending failure.

©Cummins Inc



01r00106

Figure 33: Deep wear on a connecting rod trimetal upper bearing shell without nickel layer.

Figure 33 is a close up of one of the upper rod bearing shells from Figure 32.

This bearing does **not** have a nickel layer.

Note that the copper lining does **not** have any cracking of the lining or indications of dirt or debris in the oil. There is no indication of excessive heat.

This bearing is worn but has **not** failed.

The dark staining is most likely caused by excessively long oil change intervals.

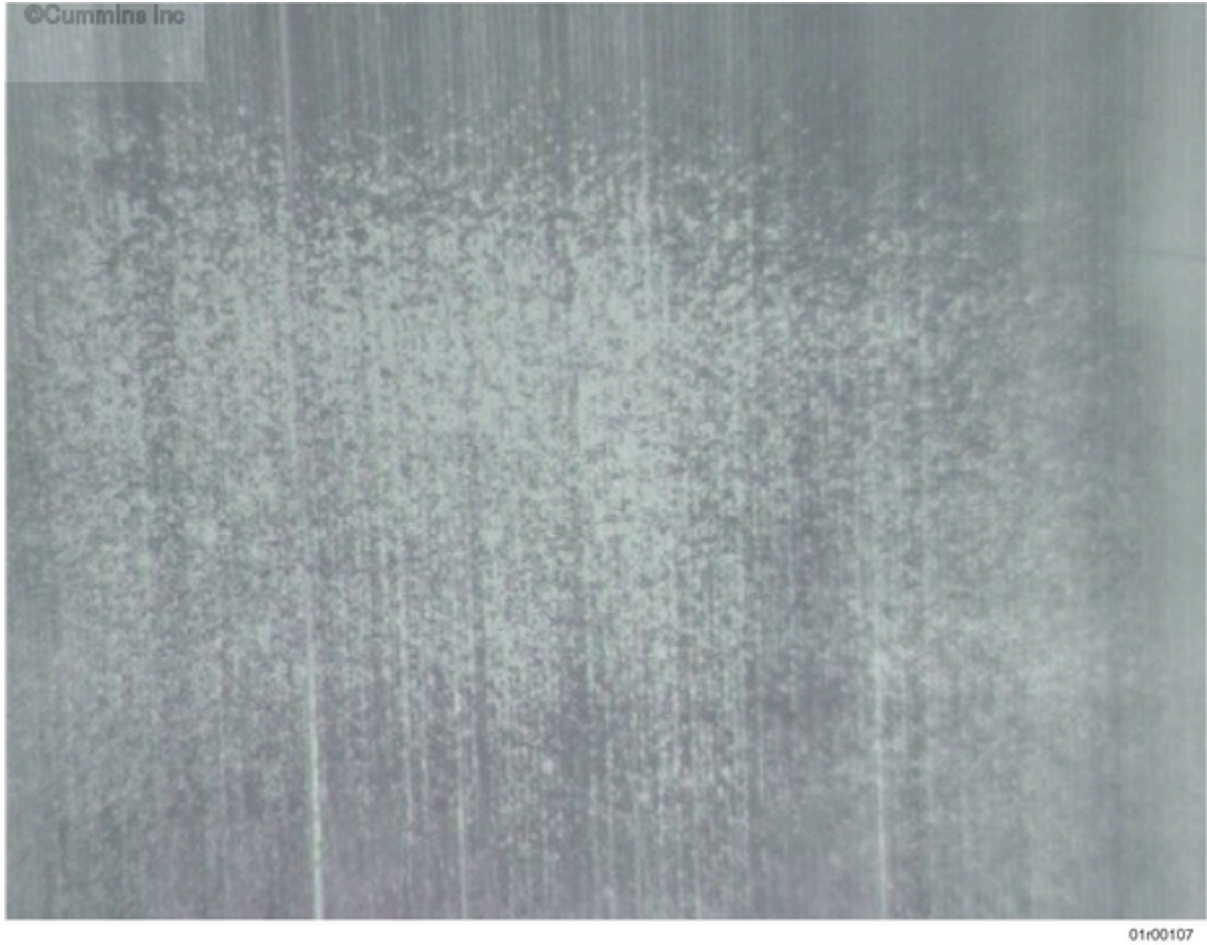


Figure 34: Most common form of overlay fatigue on a trimetal rod bearing shell.



Figure 35: A less common form of overlay fatigue on a trimetal rod bearing shell.

With overlay fatigue, the surface of the bearing exhibits cracking and/or loss of overlay material. Bearing overlay fatigue occurs when the cyclic loading exceeds that which the overlay material is able to carry. The loss of overlay material is typically confined to a relatively small, less than 15 percent, of the total shell surface area.

Overlay fatigue is typically found **only** on upper rod bearing shells and lower main bearing shells.

This is a normal occurrence during the life of most tri-metal bearings in highly loaded engines.

Figure 34 shows the most common form of overlay fatigue.

Typically, there will be little to no indication of the displaced overlay material in the lower rod bearing shell or upper main bearing shell.

What is shown in Figure 34 is **not** a failure, unless it occurs at low miles or hours.

When this occurs rapidly at early hours or miles, there are several possible causes:

- Load concentrations due to irregular loading, debris contamination, poor shaft geometry, or mis-assembly
- Material strength reduced by high temperature operation or corrosion
- Shell flexing due to insufficient crush
- Quality issue.

Figure 35 shows a less common form of overlay fatigue. Note the “tree branch” sort of overlay loss as opposed to the dot type overlay loss in Figure 34.

This sort of fatigue most often occurs at low miles or hours.

This sort of fatigue is most often the result of a quality issue and often will have overlay material in the lower rod bearing or upper main bearing, near the parting line of the two bearings in the direction of rotation of the shaft.

This sort of overlay fatigue becomes an issue when it covers a significant, more than 50 percent, of the total bearing surface area and leads to cracks in the nickel layer that may progress into the lining.

Note : Bimetal bearings do not have an overlay layer so are not subject to overlay fatigue. See Lining Fatigue.

Lining Fatigue - Trimetal Bearing:

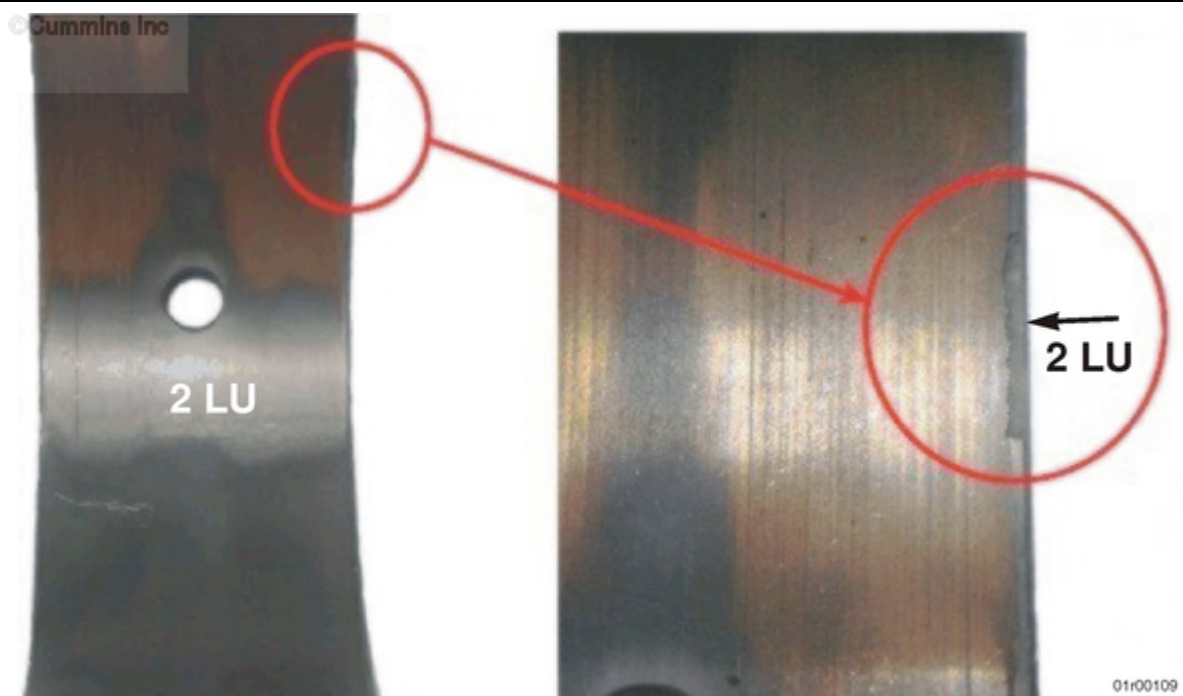


Figure 36: Lining fatigue of a trimetal rod bearing upper shell.

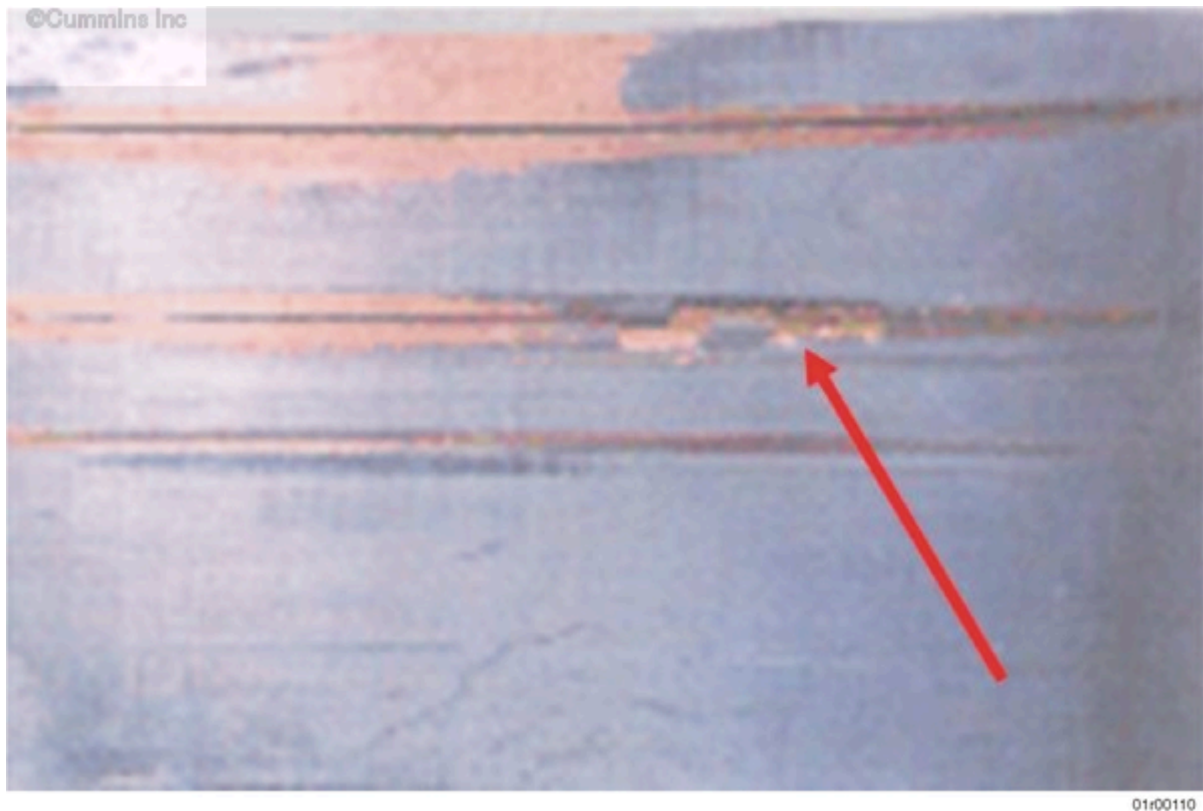


Figure 37: Lining fatigue of a trimetal bearing shell.

The surface of the bearing exhibits cracking and/or loss of lining material. Bearing lining fatigue occurs when the cyclic loading exceeds that which the lining material is able to carry. There are several possible causes:

- Load concentrations due to irregular loading, debris contamination, poor shaft geometry, or mis-assembly
- Poor contact or support provided by the bearing housing
- Material strength reduced by high temperature operation or corrosion
- Flexure and/or poor geometry of housing or shaft
- Low crush due to oversize bore or undersize shaft.

Fatigue cracks generally propagate in an approximately perpendicular direction to the bearing surface. Before reaching the bond interface, the cracks deflect, run parallel to the bond line, and join. Lining material can become detached from the bearing and removed by the oil flow.

Lining Fatigue - Bimetal Bearing:

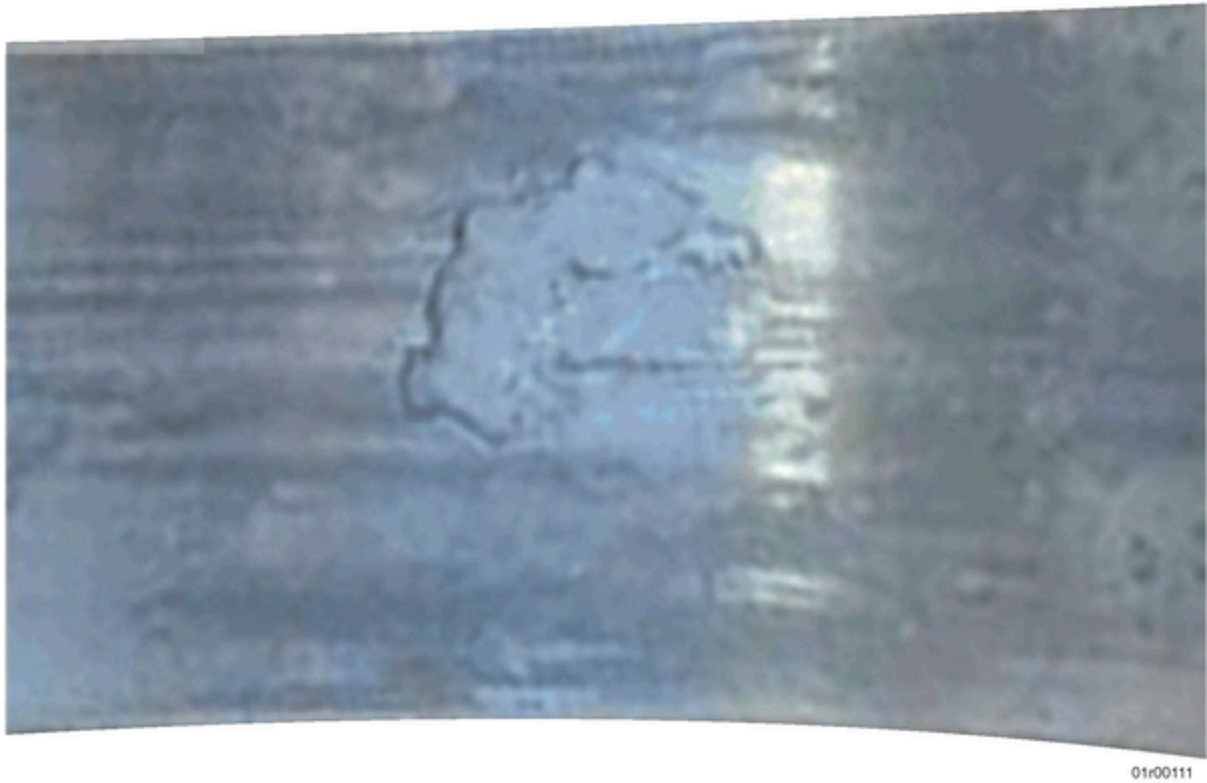


Figure 38: Lining fatigue of a bimetal lower main bearing shell.



Figure 39: Early lining fatigue of a bimetal bearing shell.

The surface of the bearing exhibits cracking and/or loss of lining material. Bearing lining fatigue occurs when the cyclic loading exceeds that which the lining material is able to carry. There are several possible causes:

- Material defects, particularly at bond with steel back
- Load concentrations due to irregular loading, debris contamination, poor shaft geometry, or mis-assembly
- Poor contact, fitment, or support provided by the bearing housing
- Material strength reduced by high temperature operation or corrosion
- Flexure and/or poor geometry of housing or shaft
- Excessive clearance with shaft
- Low crush due to oversize bore or undersize shaft.

Fatigue cracks generally propagate in an approximately perpendicular direction to the bearing surface. Before reaching the bond interface, the cracks deflect, run parallel to the bond line, and join. Lining material can become detached from the bearing, and removed by the oil flow.

Corrosion:

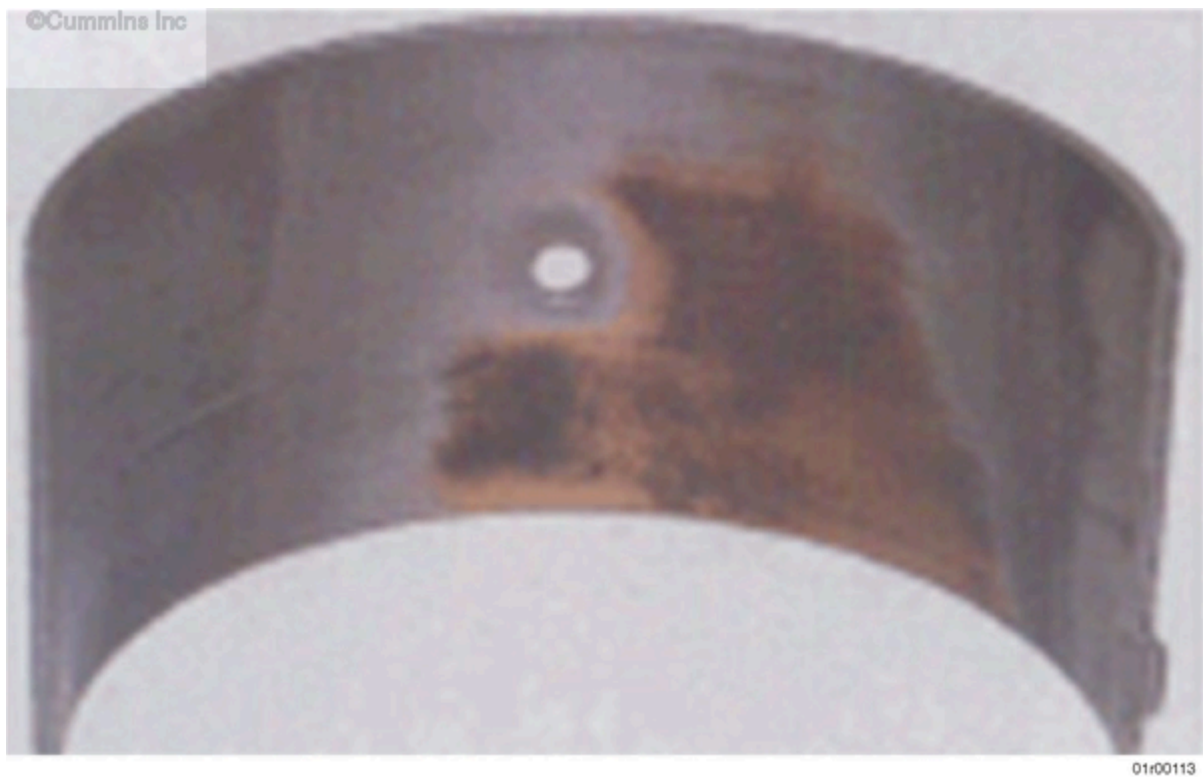


Figure 40: Corrosion of a trimetal rod bearing shell.



Figure 41: Corrosion of a trimetal bearing shell that has been exposed to glycol in the coolant.

Corrosion arises from the chemical or electrochemical reaction between a metal and substances with which it comes into contact, i.e. electrolyte solutions, humidity, or gases.

Mechanical loads can also favor corrosion. The consequences are a loss of elements within the bearing material alloy, pitting corrosion, cracks, and rust. Corrosion is recognized to increase at an increasing rate with temperature.

In an engine, corrosion occurs due to chemical attack of the bearing alloy by compounds in the lubricant. These compounds may be foreign to the lubrication system, as in the case of water or engine coolant, or may develop during operation as a result of oxidation of the oil due to increased oil change intervals or high temperature operation. The damage that takes place when a bearing operates in a corrosive environment may be the direct removal of one or more of the alloying elements or the formation of a brittle oxide over the bearing surface. In the former case, the attacked metal is removed from the alloy matrix. The weakened structure can **not** carry the load and large sections of the lining material can be removed by fatigue. A brittle oxide film is likewise a weak material. Consequently, a bearing whose surface is oxidized is more susceptible to removal of the lining through fatigue than the original bearing.

The bearing may exhibit a blackened pitted appearance in the bore. In the case of a tri-metal bearing, the depth of penetration can be isolated to the overlay or if the bearing system is allowed to operate under corrosive conditions for prolonged periods, corrosion can start to attack areas of exposed substrate.

Aluminum bimetal materials are generally resistant to corrosion in engine oil. If corrosion is present it will appear as white particles on the metal which will leave a small pit when wiped off. If corrosion is present on a bimetal bearing it is most likely caused by an engine being out of service for an extended period without proper storage.

When used, the nickel layer of a trimetal bearing provides good corrosion resistance. However, if the wear or debris damage penetrates the nickel layer and exposes the copper-lead lining, the corrosion will begin to attack the lead in the lining.

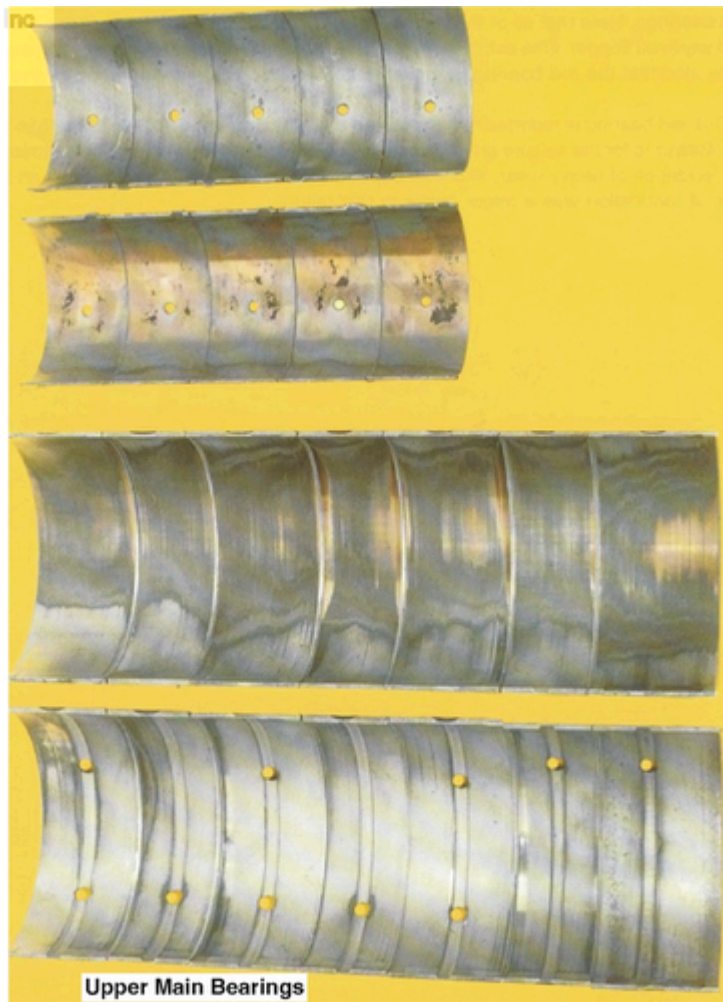


Figure 42: Corrosion of the lining of bimetal bearings without a nickel layer.

Figure 42 shows a set of trimetal main and rod bearings that exhibit a typical pattern for corrosive attack on the lead contained in the copper-lead lining. While this set of bearings does **not** have the nickel layer between the overlay and the lining, the appearance of the lining would be the same if the wear had gone through the nickel layer.

Note : The appearance shown in Figure 42 does not apply to bimetal bearings.

The brownish-orange coloration of the exposed copper areas in the lower main bearings is indicative of corrosion also.

The bearings from one of the rods seized, was destroyed, and are missing from the set. The reason for the seizure is **not** absolute since that pair of bearings can **not** be examined. However, the degree of wear and corrosive damage to the other upper rod bearings indicate lack of effective lubrication and corrosion were major factors in this failure.

This failure is **not** a failure of the bearings.

A criterion for determining when the lubricating oil has been used too long or has an inadequate additive package is that the acidity (TAN, Total Acid Number) is excessive. Excessive acidity results in a corrosive attack on the metals in the bearings. Deep wear that exposes the copper-lead lining will increase the bearings' vulnerability to corrosive attack. With the more corrosion resistant overlay and nickel layer removed, the acid in the lubricating will attack the pure lead in

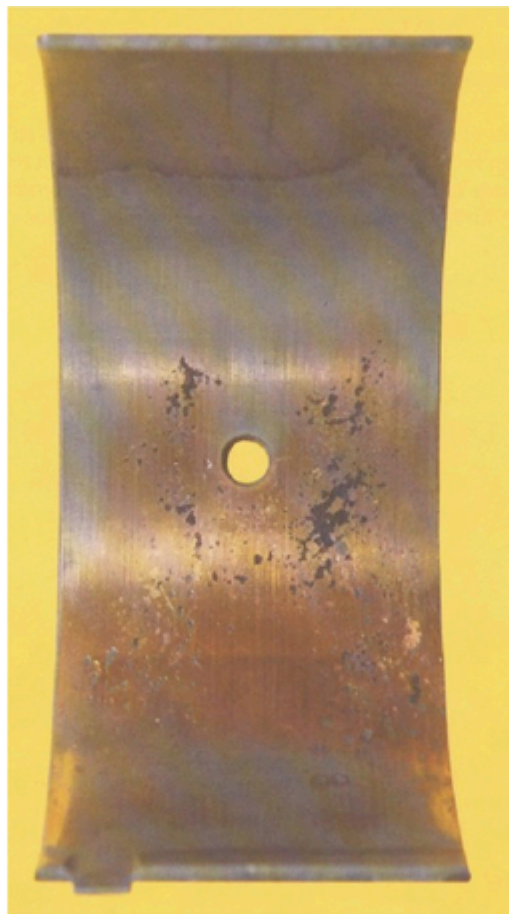
the copper-lead matrix and remove it. The honeycomb structure of the remaining copper lining has a very much reduced fatigue strength. The bearing lining will then crack and flake out in patterns very similar to fatigue damage as shown earlier.

Sometimes a fatigue damage pattern is confused with corrosion damage. Corrosion damage can be identified by a slightly different pitting pattern than that of fatigue. Usually, pitting due to corrosion is more numerous and smaller than fatigue and will be darker in color. The exposed copper will have a deeper orange-to-brownish color when corrosion is present.

In these cases, an analysis of the engine oil may provide key information.

Remember that the bearings will **not** corrode unless there is something wrong with the lubricating oil. Either the oil change interval is too long or there is a problem with a power cylinder or fuel system component causing the acidity to be greater than it should.

©Cummins Inc



01r00116

Figure 43: Close up of an upper rod bearing with corrosion damage.

Figure 43 is a close up of one of the upper rod bearings from the set in Figure 42.

The lead in the copper-lead lining layer has been removed by corrosion. The corrosion is due to excessive acidity in the engine oil along with deep wear that has removed the overlay layer.

Note : The appearance shown in Figure 42 does not apply to bimetal bearings.

Notice how the pits are more rounded and deeper than is seen with fatigue damage.

Lack of Lubrication:

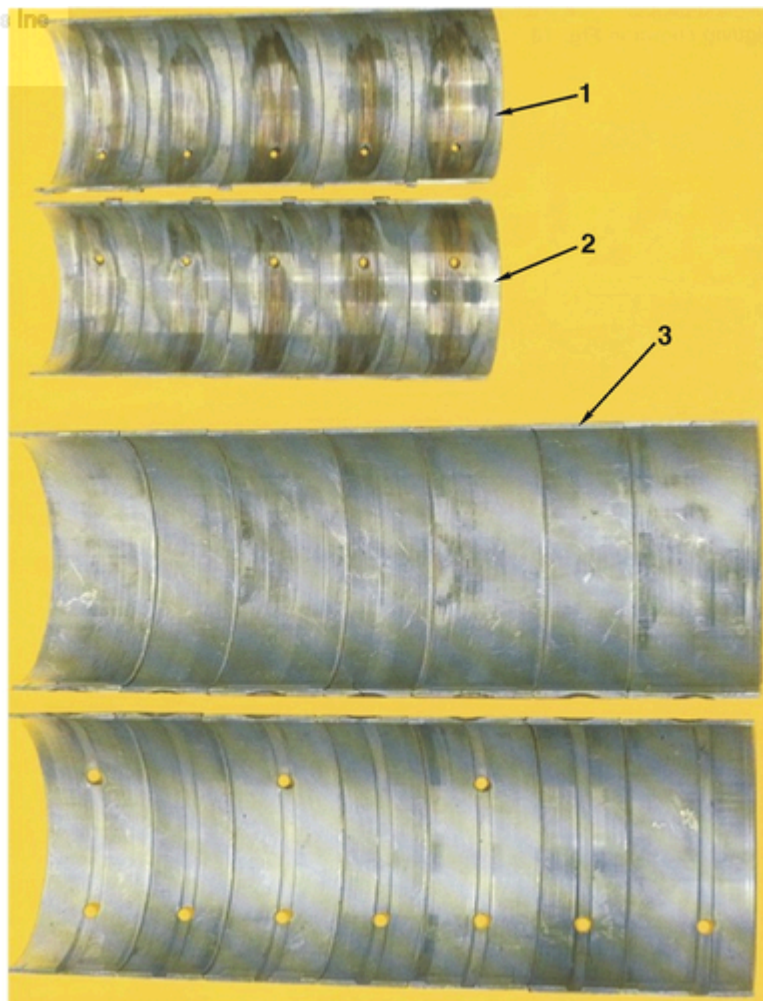


Figure 44: Lack of lubrication on a set of trimetal bearings.

The bearing damage patterns discussed up to this point result from minor lack of effective lubrication or very short periods of no lubrication. When the lack of effective lubrication is more substantial or lasting for a longer time, bearing surface temperatures rise dramatically. Expansions and distortions of the bearings and the journal occur with increased temperature. Bearing to shaft clearances are greatly reduced. The increased metal-to-metal contact along with the rising temperature burn away any lubricant is left in the journal so the temperatures are further elevated. Seizure then takes place, tearing the bearing surface and/or allowing the bearing to spin in the bore. In some instances the bearings can be completely destroyed.

There can be wide variation in bearing damage patterns resulting from lack of effective lubrication. Because of the temperature compounding effect, there is sensitivity to slight variations in clearances and the amount of lubrication reaching the bearing. In the initial stages of lack of effective lubrication, the elevated temperature will cause the materials used in the bearing overlay (if present) and lining to melt. The melting material smears over the surface and partially serves as a lubricant and heat sink, retarding further temperature rise for a relatively small period of time. As the soft metals are burned or carried away, the destruction progresses much more rapidly.

The bearing damage patterns observed due to lack of effective lubrication are dependent upon:

- How the lack of lubrication occurred in terms of the time (instant or prolonged)

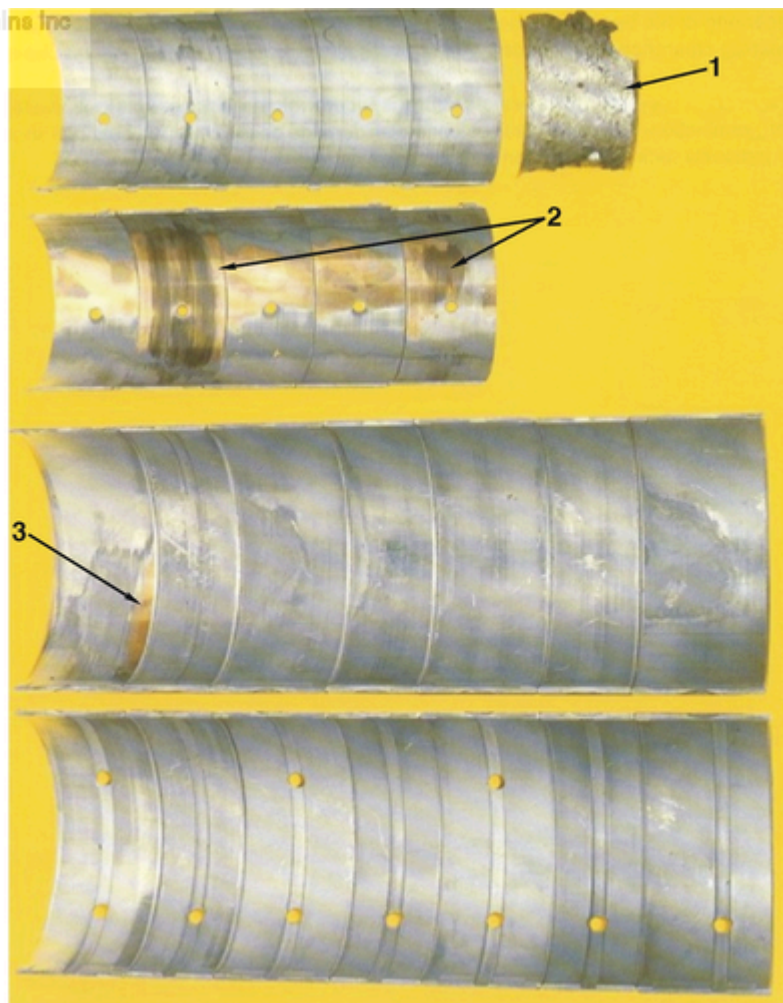
- How the engine was operated during the damaging conditions in terms of engine speed and load (power output)
- How far the damage had progressed before the engine was stopped or the damaging conditions ceased (did the operator turn the engine off or did the engine continue to run until the parts were locked together causing a sudden stop).
- How far out of alignment the shaft bent due to heat.

Figure 44 displays a typical lack of lubrication failure of the trimetal upper rod bearings (1), and the lower rod bearings (2). Note the patterns of discoloration which indicate that the bearing metal temperature has exceeded the melting point of the overlay and the lining. Clearances were reduced to a point that heavy wiping and overheating damage has progressed to the lower rod bearings which normally have much less load than the upper bearings.

One of the bearings had seized and was totally destroyed. This is **not** unusual. The remaining rod bearings show fairly uniform damage but **not** exactly the same degree of damage. Again, this is **not** unusual. The degree of visible damage can vary greatly in the same engine.

The fact that the main damage show very little to no wiping damage indicate that there was some oil flow and that it was sufficient for the lower main bearings in particular. However, there was **not** sufficient oil flow for proper lubrication of the upper rod bearings in particular. The rod bearings are exposed to higher loads. The rod bearings also have less oil film thickness because they are at the end of the oil flow.

Figure 44 is a classic example of an engine running out of oil at high load and/or high engine speed.



01K00118

Figure 45: Lack of lubrication failure on a set of trimetal bearings.

Figure 45 illustrates the variation in appearance in bearings from an engine suffering a lack of lubrication failure.

The **only** remains of two bearings in one rod is a portion of the steel backing of one shell (1). The overlay and lining have been torn and worn away as the bearing spun in the bore of the rod. Close examination of the marks on the part will show that it has been ground away. It is unknown if this portion of the back is from an upper or lower bearing shell.

Two upper bearings (2), show the characteristic discoloration and heavy wiping caused by lack of effective lubrication. Note that the damage on the bearing to the left has progressed more than the bearing to the right. The bearing to the left was hot enough to close the clearance as evidenced by the seizure wiping to be present in the lower shells.

Since the bearing (1) is so badly destroyed, it is impossible to absolutely determine why it failed. The condition of the other rod bearings indicates that a lack of lubrication is the most likely cause for the bearing failure. Given the deep wear on the other rod bearings it is also likely that these bearings have been subjected to a lack of effective lubrication more than once, but it is **not** possible to conclusively say how many times.

The area overlay removal and copper exposure to the one lower main bearing (3), is most likely the result of the crankshaft centerline mis-alignment caused by excessive heat on the rod journal. These bearings were **not** marked for location when received. The damage to the lower main bearing (3) then is progressive damage and **not** a precursor to the rod bearing damage.



01K00119

Figure 46: Set of trimetal bearings from a destructive test.

Figure 46 shows a set of bearings from an engine that had the oil drained in a test cell.

The engine was built with new, standard, trimetal main and rod bearings and installed in a test cell. The engine was started and warmed until the coolant temperature reached the thermostat opening temperature. The engine was then set to full speed and load, (350 hp @ 2100 rpm) and allowed to run until the load, temperatures, and pressures to stabilize.

The oil drain plug was then removed from the oil pan allowing the oil to drain from the pan.

After about six minutes the oil pressure began to fluctuate, an indication that air was being drawn into the lubricating oil pump. Very soon thereafter the engine speed began to drop indicating an increase in friction, most likely from the crankshaft shaft and/or rod bearings beginning to seize.

In the case of the bearings shown in Figure 46, the test cell operator decreased the load in effort to keep the engine speed at 2100 rpm. The engine speed quickly began to decrease so the engine was stopped.

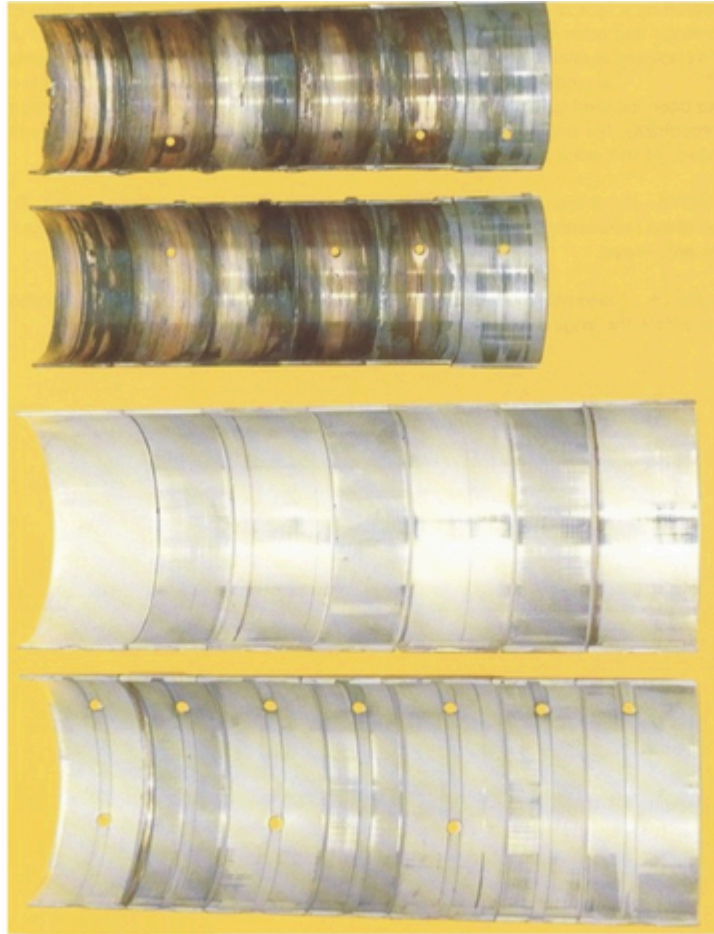
The time from the first sign of low oil pressure to the time then engine was stopped was less than one minute.

The number 3 rod bearing (third from the left) was completely destroyed. The number 4 rod bearing, (fourth from the left), was nearly destroyed.

Note that **all** the other rod bearings indicate severe overheating and melted material.

Note that the main bearings show some wiping of the overlay. The number 1 main bearings (on the far left), have copper showing on the edge. The wiping and the copper exposure are **all** the result of the crankshaft main and rod journals moving out of correct alignment due to overheating from the friction that results from the bearing seizure.

©Cummins Inc



01:00120

Figure 47: Set of trimetal bearings from a destructive test.

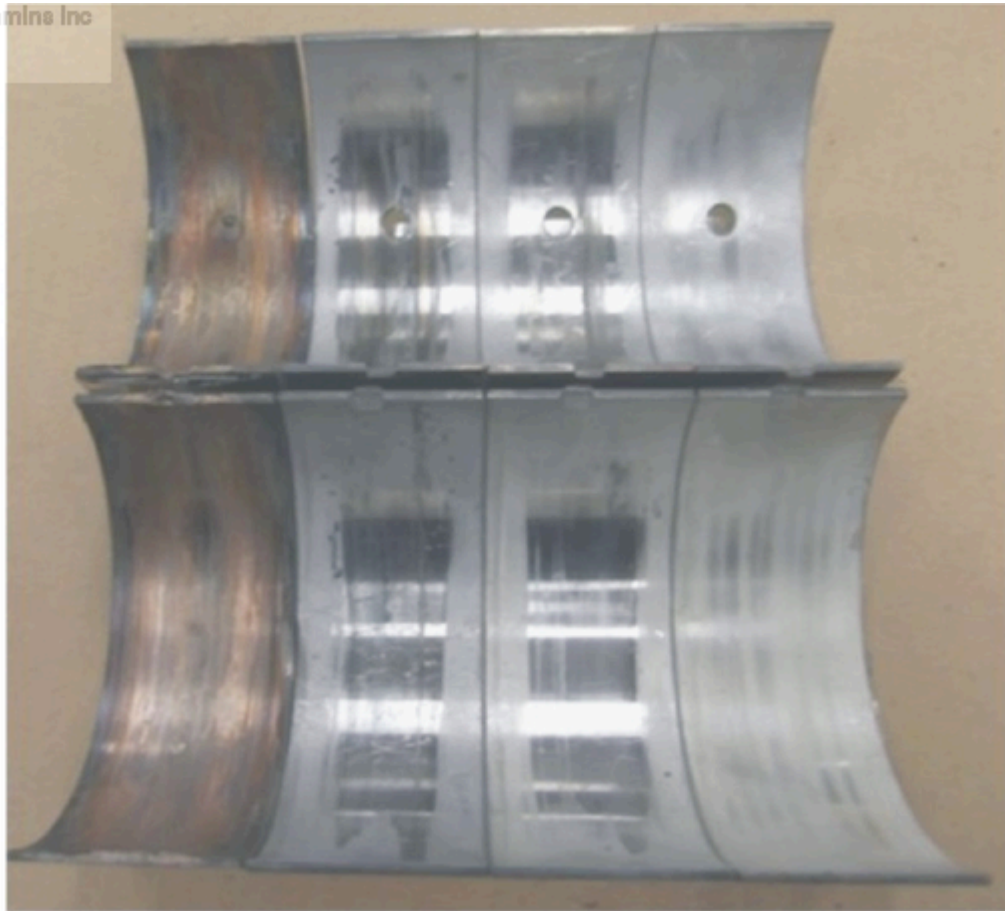
Figure 47 shows a set of bearings subjected to the same preparation and test as the bearings shown in Figure 45.

However, as soon as the engine speed began to drop because the rod bearings were beginning to seize the engine was stopped.

Note that **all** the rod bearings show damage from being overheated so the overlay and lining were melting. However, they are **not** as extensively damaged as the bearings in Figure 46.

Also note there is much less wiping of the overlay on the lower main bearings.

The difference in color in the main bearings in Figures 46 and 47 is due to the lighting used when the photographs were taken. The bearings are the same part number and construction.



01r00121

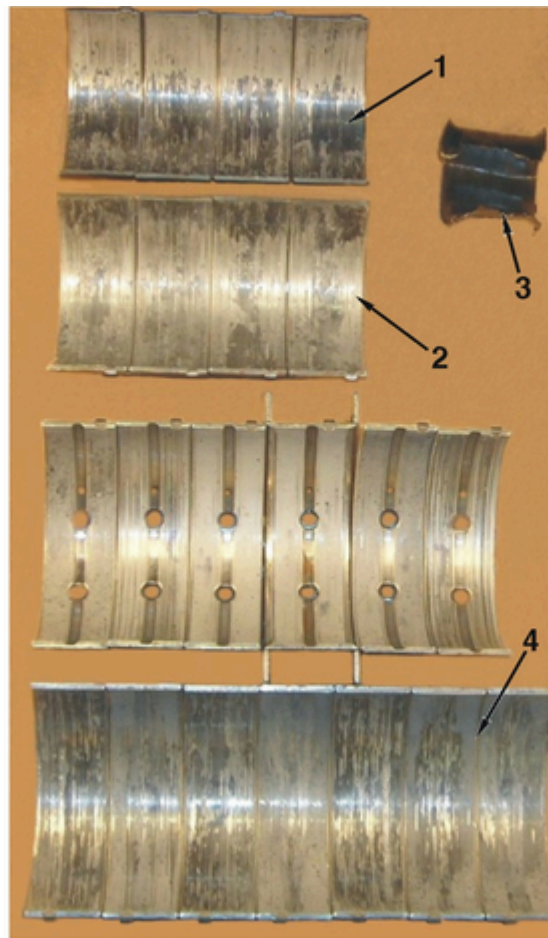
Figure 48: Variation of bearing damage from a lack of lubrication failure.

Figure 48 shows four pairs of rod bearings that suffered a lubricating oil pump failure at two hours in at the customer.

Note the great variation in the degree of damage.

These bearings have a nickel layer between the overlay and the lining. Note the wiping of the overlay to the nickel layer in the four center bearings.

These are the **only** bearings returned from the engine failure and were **not** marked for cylinder location.



01r00122

Figure 49: Partial set of bimetal bearings exhibiting a lack of lube failure.

Figure 49 shows a set of bimetal bearings exhibiting a lack of lubrication failure.

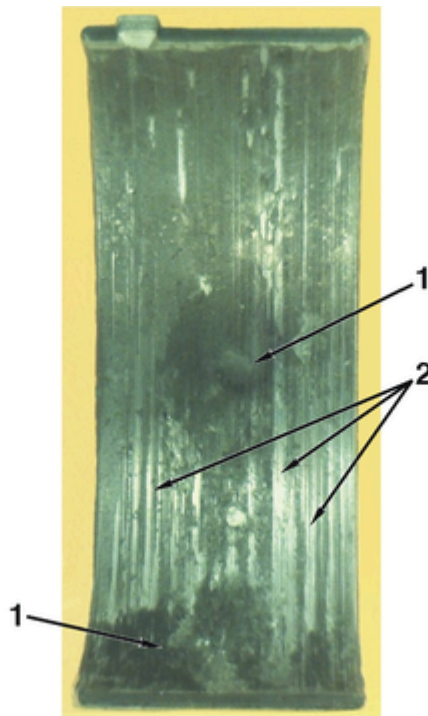
Figure 49 is of a partial set of bimetal bearings from a B5.9 engine that suffered a lack of lubrication failure.

Note how the lining material has been torn away on the upper rod bearings (1), with the material transferred to the lower rod bearings (2). Note that none of these bearings have spun in the rod bore.

The dark areas in the center of the upper rod bearings (1), and the upper edges of the lower rod bearings (2), are actually smeared lining material. This can be easily confused with the steel back.

One rod bearing (3), has been completely destroyed. It has been further damaged after coming out of the rod as evidenced by the shape. The rod cap came off the connecting rod holding this bearing due to the rod bolts being overheated and yielding in tension.

Note also how the material has been torn away from the lower main bearings (4) with some material transfer to the upper main bearings.



01r00123

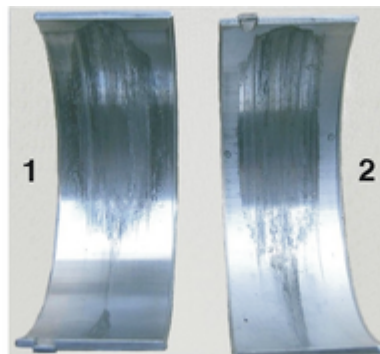
Figure 50: Bimetal upper rod bearing from a lack of lubrication failure.

Figure 50 is a close-up of one of the upper rod bearings from Figure 49.

The dark areas (1) are smeared and burned lining material.

The marks (2) are actually very deep gouges in the lining material that do **not** quite reach the depth of the steel back.

This bearing shell does **not** have an oil hole.



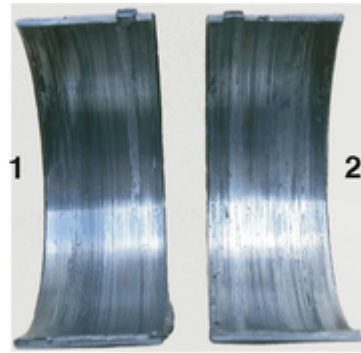
01r00124

Figure 51: Bimetal rod bearings lack of lubrication failure.

Figure 51 shows bimetal upper (1) and lower (2) connecting rod bearings from a B4.5 engine at low hours.

The damage is the result of a lack of lubrication that has **not** progressed as far as the damage shown in Figures 49 and 50.

Again, the dark areas are where the lining material has been overheated and smeared.



01r00125

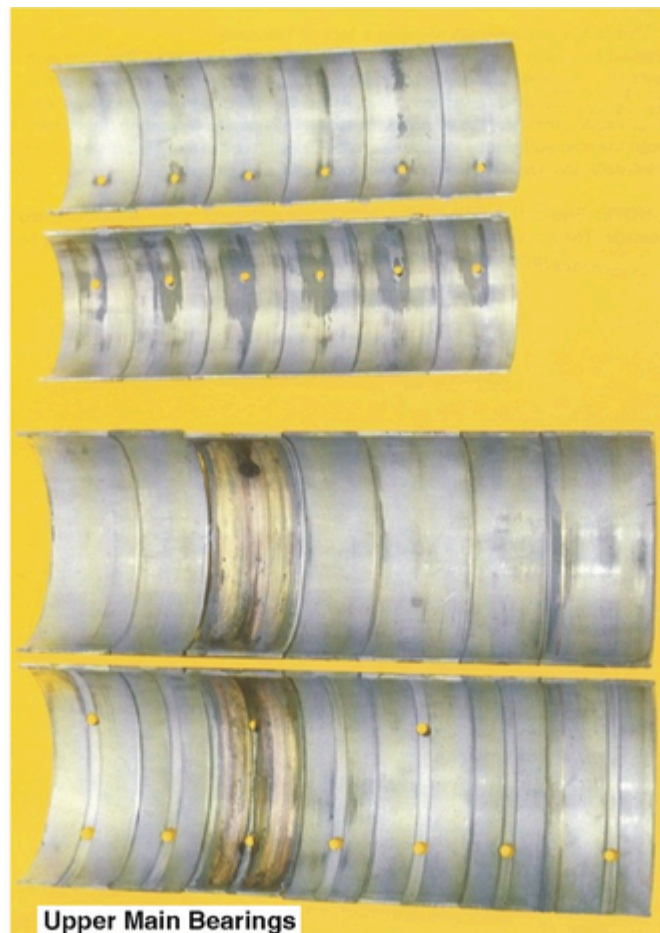
Figure 52: Bimetal rod bearings lack of lube failure.

Figure 52 shows the upper (1) and lower (2) rod bearing out of the same engine as the bearings in Figure 51.

Note how the damage has progressed from Figure 51. Also note that the bearings have not spun.

One of the rod bearings in the engine was completely destroyed and the other three exhibit damage from lack of lubrication.

Cold Start/Fast Start Main Bearing Seizure:



01r00126

Figure 53: Trimetal bearings subjected to cold start.

Figure 53 shows a set of trimetal main and rod bearings that have been subjected to cold start/fast start.

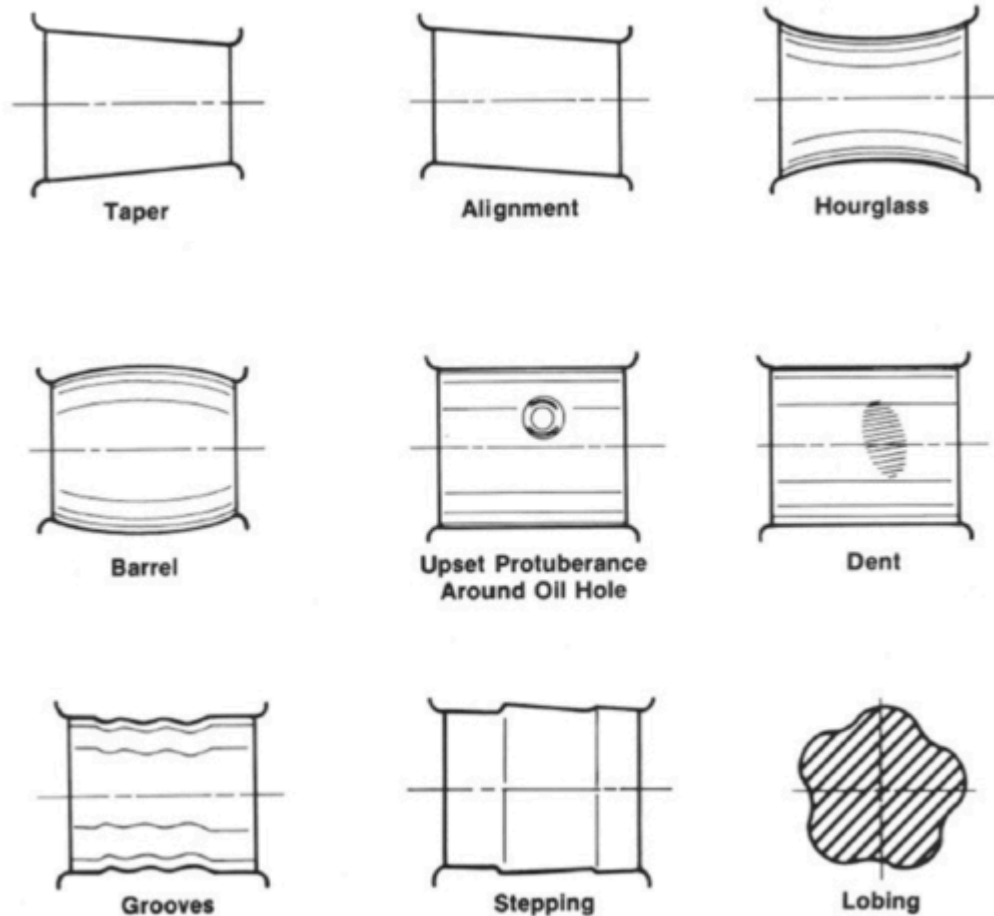
This type failure is usually encountered in cold weather when the then is started with uncontrolled ether injection, although it can also occur when ether is **not** used. The seizure is caused by operating the engine at high speed immediately after starting, before adequate oil flow has been established.

The usual sequence of events for this type of failure is that the engine has **not** been running and has cold soaked for over four hours in temperatures below 1° C [30°F]. The lubricating oil becomes more viscous at these temperatures. The engine is hard to start so the operator uses ether from a spray can to start the engine and it is immediately sped up to high idle rpm and held there. After some amount of time the engine speed begins to decrease as the ether is completely burned. Sometimes the engine will stop and sometimes the operator will continue to spray ether in effort to keep the engine running. If the engine stops there is a few seconds delay as the operator adds more ether and starts the engine. Sometimes at this point the engine will **not** crank or cranks very slowly because the bearings have started to close on the crankshaft due to heat from shaft to bearing contact. During the high-speed operation, one or more main bearings can be starved for oil and overheat to the point the bearing lining material melts. Since the condition occurs before lubricating oil flow is stabilized, usually one or more main bearings will be affected, although the rest of the bearings can show some degree of wiping. When the engine stops, the heat-sink effect of the cold crankshaft quickly solidifies the molten bearing material, freezing the bearing to the crankshaft journal.

Note that in Figure 53, **only** one main bearing, Number 3, has seized. The upper rod bearings show some wiping but otherwise **not** further distress appears. Upon close examination of the seized main bearings, beads of solidified bearing material are evident along the edges of the bearings. In some cases, a few drips of the solidified material can be found in the bottom of the oil pan.

There are some variations of in this failure pattern. In some situations, repeated attempts to start the engine after the seizure will cause the bearings to spin in the bore.

Geometric Deficiencies:

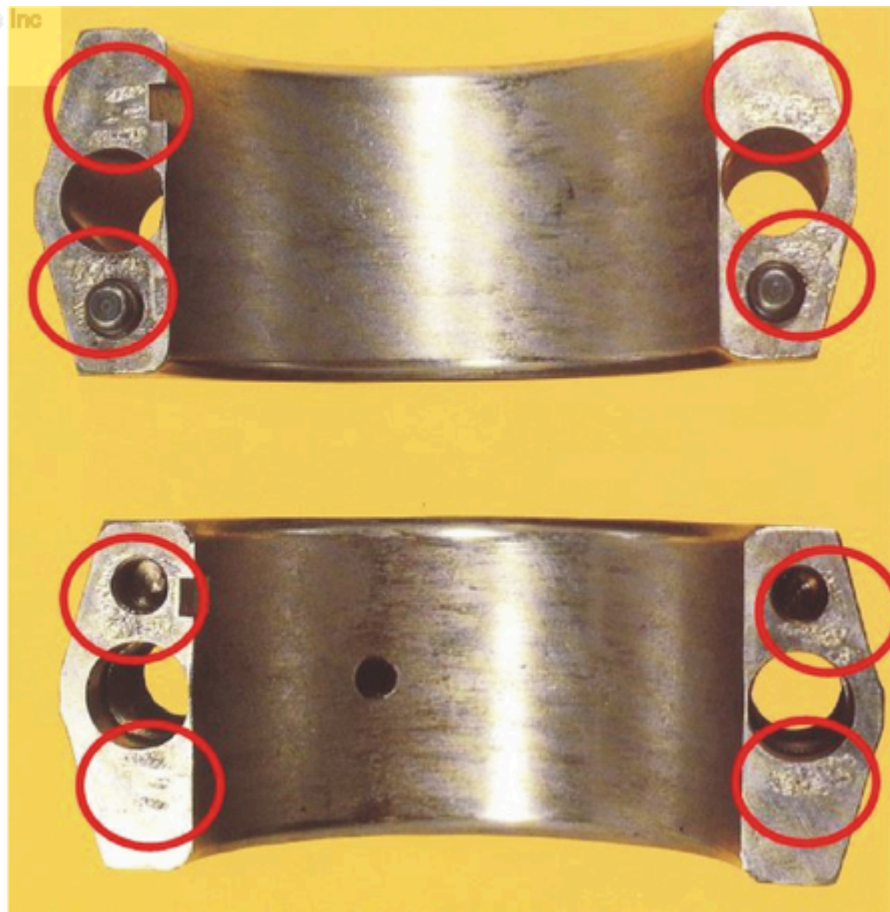


0100127

Figure 54: Geometric deficiencies in shafts and bearing bores.

Figure 54 depicts examples of geometric deficiencies in shafts and bores that can effect bearing performance. These deficiencies are caused by incorrect machining or reconditioning of shafts and bores. Any of these deficiencies will affect the lubricating oil film on the bearing and shaft. The amount of the deficiency will determine whether the shaft and bearing make contact creating heat that can damage the bearing overlay and/or lining material. Insufficient clearance limits the lubricant flow so that frictional heat is **not** carried away satisfactorily, increasing the probability of overheating distortion and seizure. Excessive clearances will result in less arc of contact between the bearing and the journal greatly increasing the oil film pressure and loading and bearing overlay and lining fatigue. Wear of the bearing increases the clearance and the probability of seizure. Geometric deviations (hourglass, barrel, alignment, linearity, etc.) which cause clearances to increase the probability of metal-to-metal contact and damage patterns will be evident in the bearing that are very similar to those for lack of lubrication. Analysis of damage due to geometric deviations is a matter of finding evidence that such deviations exist. The deviation may or may **not** be accurately measured by simple instruments such as micrometers.

Incorrect Connecting Rod Bolt Clamp Load:



01r00128

Figure 55: Loose rod bolt on a saw cut rod.

The roundness of the crankpin bore of the connecting rod is dependent on the tension of the connecting rod bolts that attach the cap to the rod. Rod bolt tension is what establishes the amount of clamp load between the rod and the cap. It is necessary that the rod cap be attached with the same clamp load as when it was machined. Deviations from correct clamp load will cause the bore to be out of round and result in correct clearances. Low clamp load can also allow the cap and rod to move relative to each other and cause fretting as in seen within the red circles in Figure 55.

The clamp load can be reduced by incorrect rod bolt tension. Incorrect tension can be the result of an inaccurate torque wrench, mutilated threads on the rod or bolt, or failure to lubricate the bolt threads correctly before torquing. A burr or piece of debris sandwiched between the cap and rod parting faces can cause a distorted bore. Also, the burr or debris can eventually be flattened enough to fall out of the joint allowing clearance that can allow the rod bearings to spin.

A rod bearing failure resulting from excessive clamp load (rod bolt tension too high), is very difficult to analyze. There is **not** likely to be much evidence the over-tensioning remaining after the failure. Attempt to measure the prevailing torque on the other rod bolts in the engine. While the condition of the other bolts in **not** conclusive evidence, it is a strong indicator that mis-assembly might have caused the failure.

Bearing failure resulting from under-tensioning can be evident in different ways. Relative motion between the cap and the rod will cause fretting damage on the parting faces, which will be evident even though the rod bearing is completely destroyed in the failure. Figure 55 shows an example of connecting rod fretting due to under-tensioning. Low clamp load also results in closing in the parting line causing parting faces to act as oil scrapers and destroy oil film generation.

Note that in Figure 55 there is no evidence of overheating on the mating faces of the rod and the cap. However, the condition of the surface of the bore of the rod and cap clearly indicate that the bearings have spun in the bore. The bore surfaces also show no indication of excessive heat. The lack of heat indicates there was **not** a lot of friction as the bearings turned in the bore. The most common way for the bore to become larger is for the cap and rod to separate, in this case due to low clamp load because the bolts had loosened during engine operation.

The lack of discoloration from heat also indicates that lubricating oil was present and provided cooling to the connecting rod. This oil would be from the pressurized drilling in the crankshaft and the splash oil that in the crankcase.

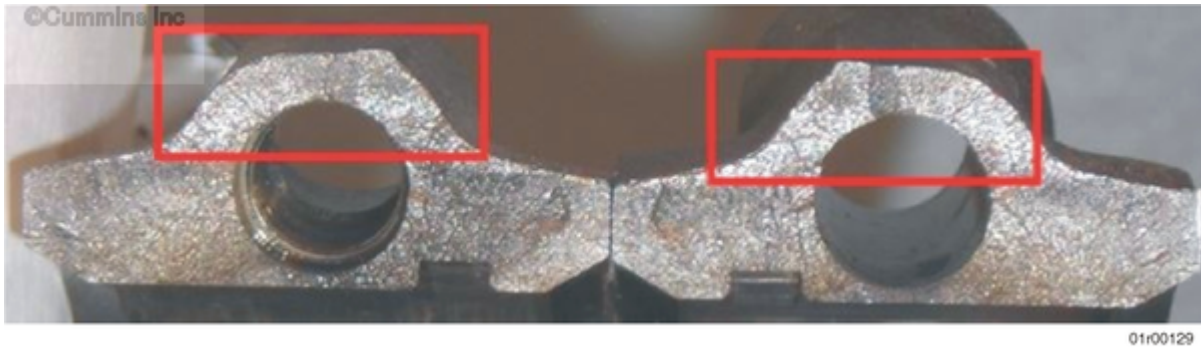


Figure 56: Loose rod bolt on a fracture split rod – tight side.

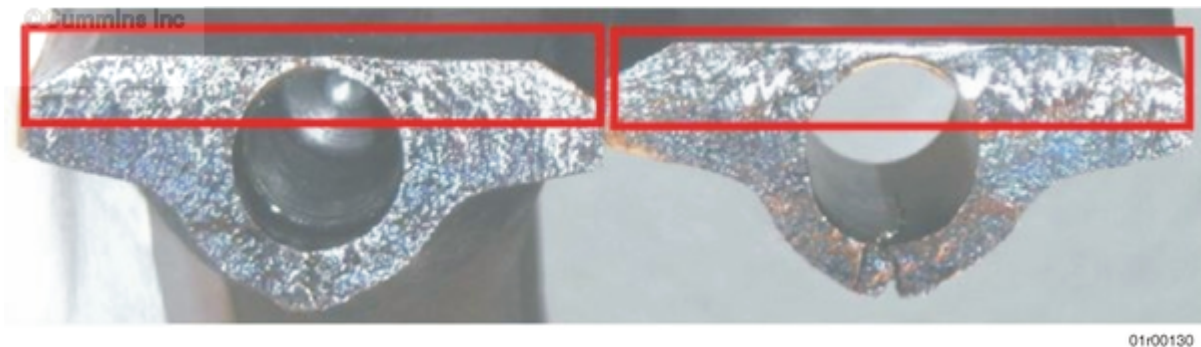


Figure 57: Loose rod bolt on a fracture split rod – loose side.



01r00131

Figure 58: Rod bearings pushed between rod and cap.

Figures 56 and 57 are of the connecting rod and cap from a B6.7 engine. The rod and cap are of the fracture split type as opposed to the saw cut type shown in Figure 55.

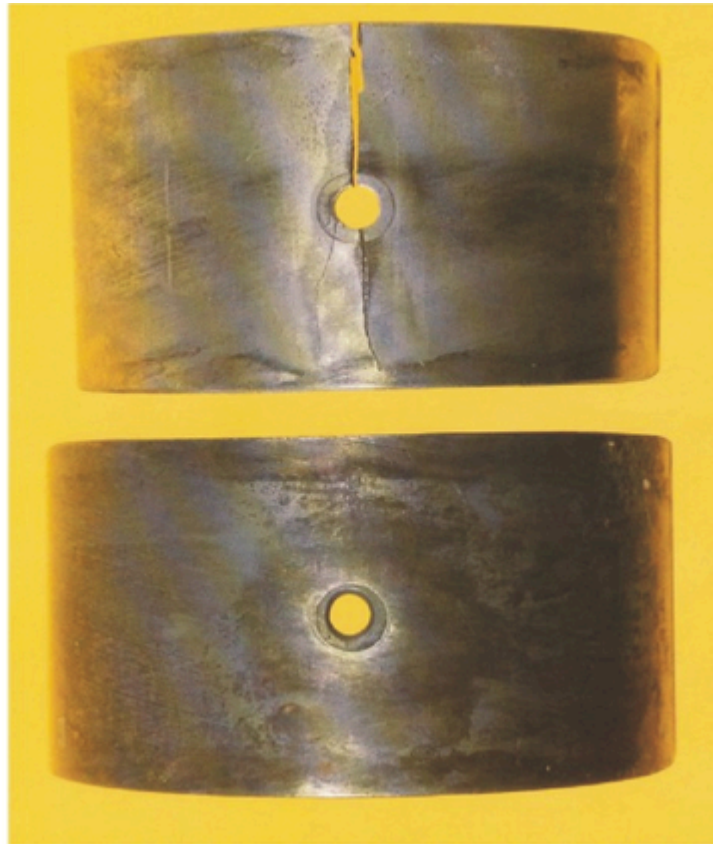
The fracture split surfaces **always** have the same rough appearance as broken iron material like that shown outside the boxes in Figure 56.

Note the shiny appearance inside the boxes in Figure 57. The shiny appearance is caused by movement between the rod and cap. The movement is allowed by the low clamp load of the loose rod bolt. In this case the rod bolt loosened enough to allow the rod bearings to be forced into the gap between the rod and the cap. See figure 58.

Note the appearance of the rod and cap face inside the boxes in Figure 56. The shiny appearance indicates some movement between the rod and cap. The area inside the boxes in Figure 56 indicates little to no movement. In this case the rod bolt on the side of the parts in Figure 55 was found to be at the correct torque. However, the amount of load on the area inside the boxes was increased beyond the design limit of the clamp load so the parts moved relative to each other.

The crack in the rod cap visible in Figure 58 was caused when the rod cap cooled off and closed in after the engine was shut down. The rod bolt on that side was still in place and was tight due to the load created by the part of the bearings forced between the rod and cap.

Note that the surface areas in Figures 56 and 57 show little indication of being overheated. There is some indication of overheating in Figure 58. However, in this case there was no problem with the lubricating oil system, so there was pressurized oil available to cool the parts to some degree.



01r00132

Figure 59: Fatigue failure of steel bearing back.

In addition to causing the rod bore to be out of round, under-tensioning also results in loss of correct crush load on the bearings. Crush load will also be reduced if the rod bore is oversize or the bearings are too short. Loss of crush load allows the bearing to move and flex in the bore, resulting in a fretting or polished appearance on the back of the bearing. Flexing causes fatigue and cracking of the steel back. Figure 58 shows an example of this condition.

Fretting of the back of the bearings and fatigue cracking of the steel back can also occur when there is severe fatigue and wear of the bearing lining. In these situations, the fretting fatigue of the steel back results from the yielding of the steel back, reducing crush and then the flexing of the loosened bearing.

Note the dark stain on the back of the lower bearing in Figure 58. These dark areas are the result of micro-movement of the bearing. Note the absence of stain around the oil hole where there is chamfer on the oil hole in the rod. This condition is normal on both tri-metal and bi-metal bearings.

Assembly Errors - Main and Rod Bearing Caps:



01r00133

Figure 60: Typical connecting rod and cap markings.

Connecting rods and caps have marks to indicate parts that were machined together; thus, they **must** be installed in an engine together.

Figure 60 exhibits typical rod and cap markings. The markings are **not** standard and vary from engine model to engine model and connecting rod supplier. However, **all** rods will have some sort of marking.

Not only must the parts be installed together, the marks **must** be on the same side of the connecting rod assembly as shown in Figure 60. Some rods are machined in such a way as to prevent the parts from being installed backward.

If the parts are mixed or **not** assembled correctly, the risk of failure is high.

It is recommended that the connecting rod and piston assemblies be marked for position in the engine before disassembly. Marking the parts will aid in failure analysis.

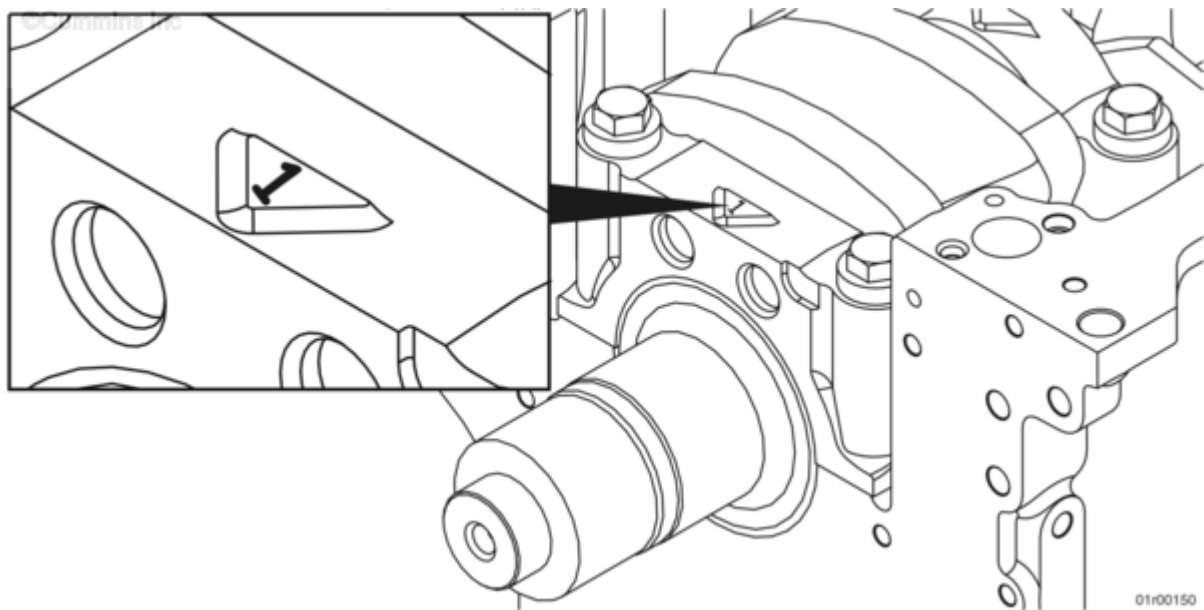


Figure 61: Make sure main bearing caps are clearly marked before removal.

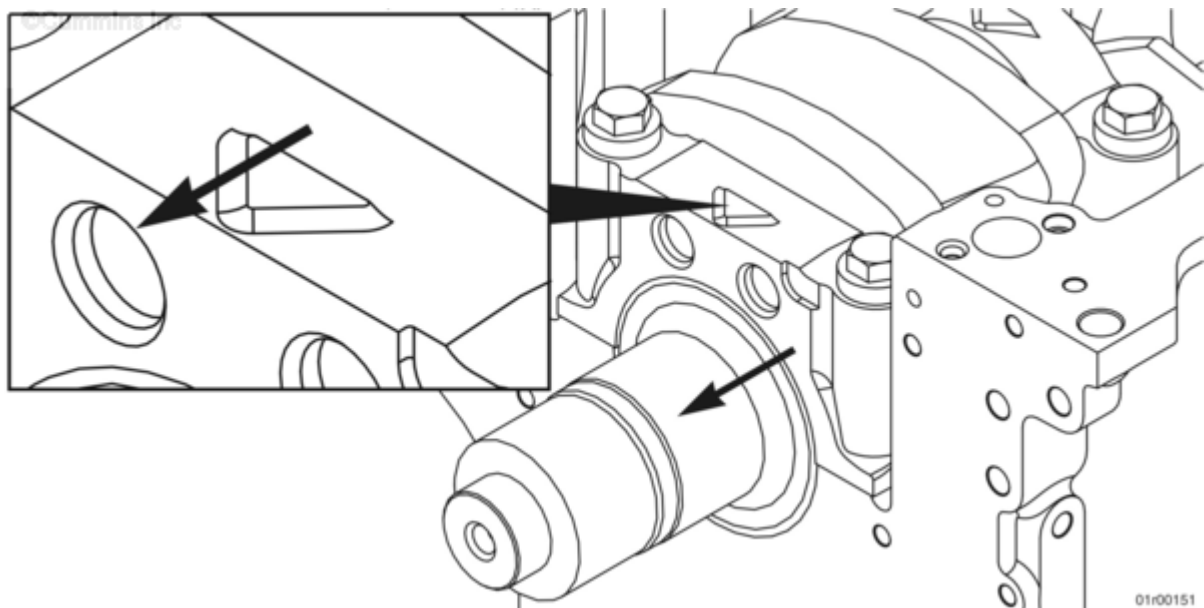


Figure 62: Main bearing caps with an arrow always point to the front of the engine.

Main bearing caps **must** always be installed as they were when the bore for the main bearing was machined. If the caps are mixed by location or orientation, a major failure is very likely. Most often, if a cap is **not** installed correctly, the crankshaft will **not** turn but this is **not always** true. A crankshaft without the connecting rods or driven members (transmissions, generators, fan drives, hydraulic pumps, lubricating oil pumps) attached, **must** be able to be turned by hand after **all** the main bearing cap bolts are tightened to specification. Engines with a displacement above 23 liters may require a short lever to turn the crankshaft, simply because of the mass of the part, but should turn freely with no tight spots.

The markings on main bearing caps vary by engine model and casting supplier.

Always make sure the location marking on the cap is legible. If **not**, mark the cap to indicate the number. Start from the front of the block with #1. See Figure 61.

Always make sure the orientation of the cap is clear before removing the cap. Some caps have a cast-in arrow to indicate the front of the engine. See Figure 62.

Make sure the main bearing caps and cylinder block are marked to indicate which caps **must** be used with which cylinder block.

©Cummins Inc



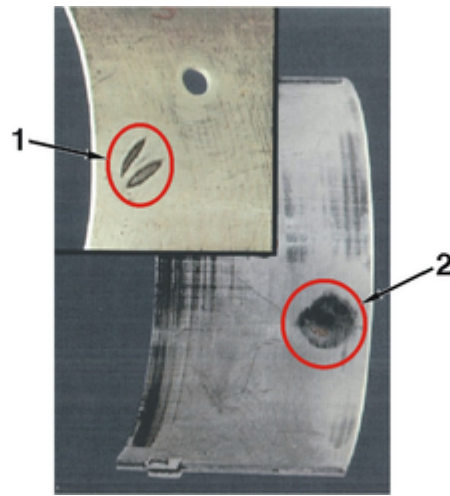
01r00134

Figure 63: Connecting rod bearing not installed correctly.

Occasionally the rod or main bearings are assembled out of position in the bore, either turned a few degrees or pushed to one side or the other. The operation of seating the cap in the bore can be blind when working on the engine installed in the machine. This is particularly true of the connecting rod bearings when working underneath the engine and the crankshaft is blocking the view. Unless extreme care and double checking are exercised, the bearings can be moved out of position and **not** noticed by the technician.

Evidence of such mis-location type failures can be found by closely examining the lock tangs on the bearings and the seating patterns of the tangs against the rod and the rod cap. The lock tangs will usually be mashed or mutilated, indicating the manner of bearing mis-location. Distortion of the bearings will result in metal-to-metal contact and heavy wear in the area of the lock tang such as shown in Figure 63.

Debris Between the Bearing and the Bore



01r00135

Figure 64: Debris between the bearing and the bore.

Debris between the bearing and the bore can cause shaft to bearing contact.

See Figure 64. Debris between the back of the bearing and the bore (1), can cause a high spot in the bearing lining (2). The result is shaft to bearing contact as shown in callout (2).

This condition increases the risk of failure due to friction and heat in the localized area. The damage can result in a single rod or main bearing failure in which the bearing is completely destroyed, leaving little or no trace of the debris in the back of the steel back. In this situation, look carefully at **all** the other bearings and bushings in the engine for signs of debris that are **not** from the bearing lining.

Loss of Bearing Crush:



01r00136

Figure 65: Bimetal bearing with fatigue from loss of crush.

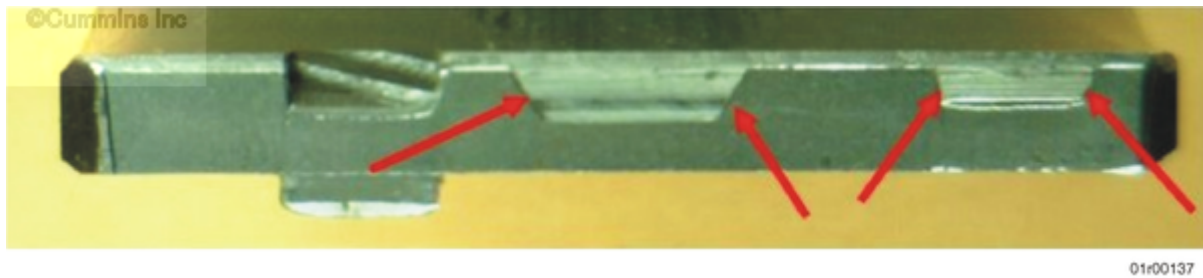


Figure 66: Close-up of Figure 65, showing bimetal bearing with loss of crush.

See the discussion on bearing crush earlier in this document. Remember that crush is what prevents the bearing from being loose in the bore. If the bearing is loose in the bore it is subject to loading that is out of design.

Figures 65 and 66 are of a bearing that had inadequate crush in the main bearing bore.

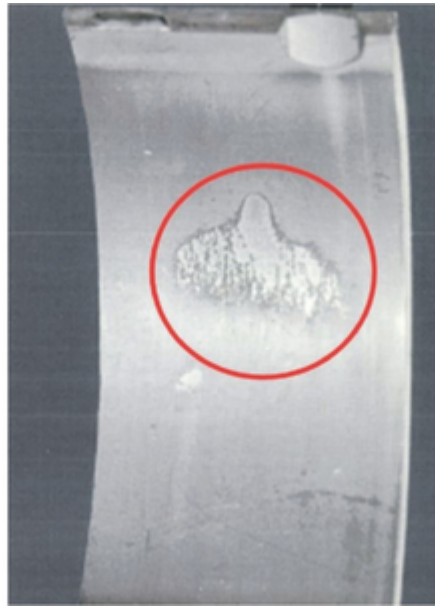
Note the loss of lining material in the center loaded area of the bearing in Figure 65. This loss of material is the result of fatigue. That is the bearing is being “hammered” as the adjacent cylinders fire.

Figure 66 is a close-up of the parting edge of the bearing. Note the arrows at the bottom of the picture. When move your fingernail across these areas, you will feel slightly raised material. In actuality, the area of the parting line **not** indicated by the arrows has been lowered by the movement between the upper and lower bearings. This is another indication that the bearing has been moving excessively in the bore. The other bearing in this bore exhibits a mirror image of the parting line.

This condition is caused by a quality issue. Either the bore is too large or one or both of the bearings are too short. Either condition can cause inadequate crush. If this condition is found, the diameter of the bore should be measured per the specific engine shop manual. It is **not** possible to accurately measure bearing length in the field as special tooling is required.

As a rough check during assembly of main and rod bearings, after they are placed in their half of the bore, feel the parting lines to make sure the bearing protrudes slightly above the mating face of the rod, rod cap, main bearing cap, and main bearing saddle. Install the bearing shell so it is even with the mating face on the bore and feel the other parting line. You should be able to feel the bearing protruding. If you can **not**, confirm the bore diameter and the bearing shell part number.

Cavitation Erosion:



01r00138

Figure 67: Cavitation erosion.

Cavitation erosion has sometimes been called cavitation corrosion. Cavitation erosion is actually a fatigue mechanism, **not** corrosion which is a chemical mechanism.

An example of cavitation erosion is shown inside the circle in Figure 67.

This erosion most commonly is **only** of a very thin layer of the bearing overlay on trimetal bearings. The erosion very rarely deep enough to reach the nickel layer or the bronze layer on bearings that do **not** have a nickel layer.

The erosion is the result of the collapsing of tiny air bubbles, cavitation, in the lubricating oil. The erosion is commonly found where the hydrodynamic wedge of oil begins to come under the heaviest load in the direction of rotation of the shaft.

If the cavitation erosion does extend deep enough on a trimetal bearing to reach the nickel or bronze layers of the bearing lining then there is likely a problem with too much air in the oil. Excessive air can be caused by operating at a low oil level or operating the engine at or over the maximum angularity allowed by the oil pan arrangement.

If cavitation erosion occurs on a bimetal bearing then there is too much air in the oil.

On both trimetal and bimetal bearings, the difference between chemical corrosion and cavitation erosion will be the color of the bearing. Cavitation erosion will leave a "clean" area and the color of the rest of the bearing will be as new. Chemical corrosion **always** causes the pits or the metal to change color. The color usually becomes darker.

Electrical Current Damage:

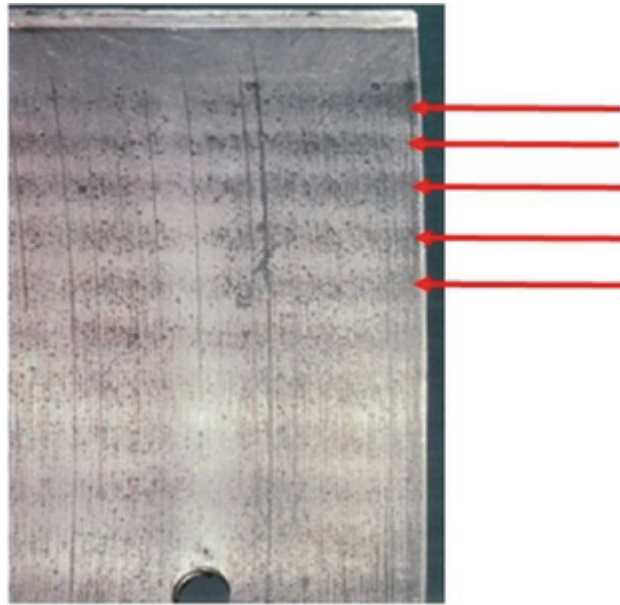


Figure 68: Electrical current damage of a trimetal bearing.

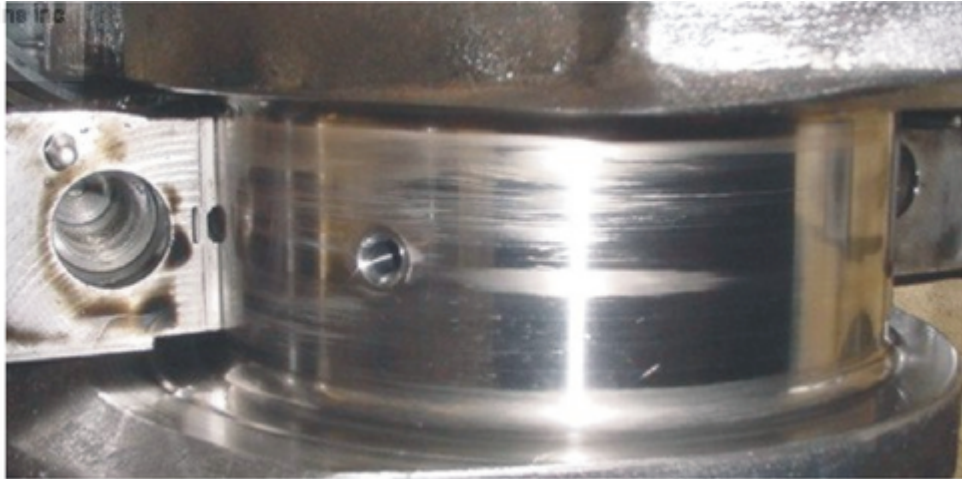


Figure 69: "Frosting" of a crankshaft main bearing journal from electrical current.

Stray electrical current through even part of an engine can result in damage to the bearings. The crankshaft bearings are more often affected than the rod bearings.

Figure 68 exhibits electrical current damage of a trimetal bearing shell. The dark areas in the "lines" of damage are actually pits where material has been removed from the bearing overlay.

Figure 69 exhibits the "frosting" of the surface of the crankshaft that may accompany bearing damage like that in Figure 68. However, the crankshaft is **not always** exhibit damage from electrical current.

Electrical current damage is nearly **always** caused by a faulty ground in the machine's electrical system or an improperly grounded electrical component on the engine. In rare cases the damage can be due to static current produced by a belt or other moving parts.

The damage will continue until the faulty electrical ground is found and corrected.

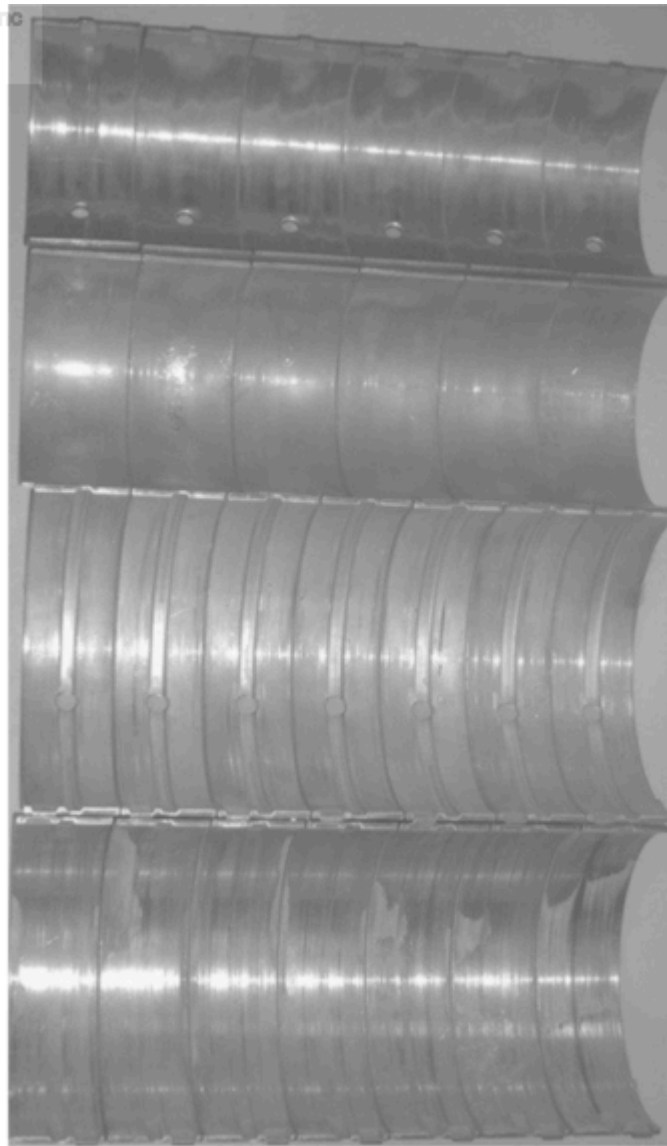
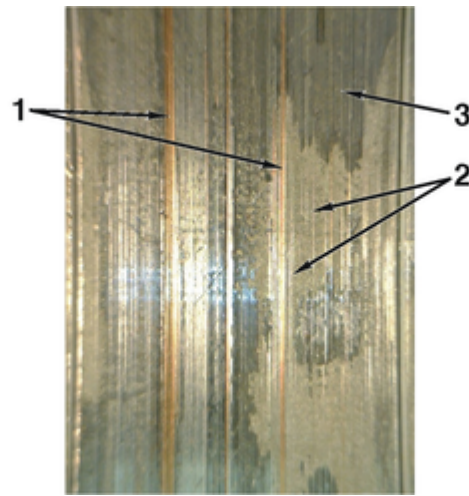


Figure 70: Large particle debris damage of trimetal bearings.

Figure 70 shows trimetal main and rod bearings from a 2010 certified X15 engine. The engine suffered a failure of the ceramic fuel pump plungers at 275,197 km [171,000 miles]. This set of bearings exhibits large particle debris damage. While these bearings are **not** reusable, there is no failure on the part of the bearings. To the contrary, the bearings exhibit good embedability performance. The hard particles of ceramic material have scratched through to the bronze lining in some areas and the bearings have been able to withstand the debris. The lubricating oil has carried the debris throughout the engine. Note that **all** of the bearings have some degree of debris damage. When **all** of the bearings indicate debris damage:

- The oil filter bypass valve has been pushed open during engine start (**all** oil filter bypass valves are opened until oil pressure is available to the clean oil side of the bypass valve).
- The oil filter has so much debris, that the filter is plugged forcing the filter bypass valve open.
- The oil filter bypass valve is stuck open, possibly by the debris from the fuel pump failure.

See the following photographs for closer views of the bearing damage.



01r00142

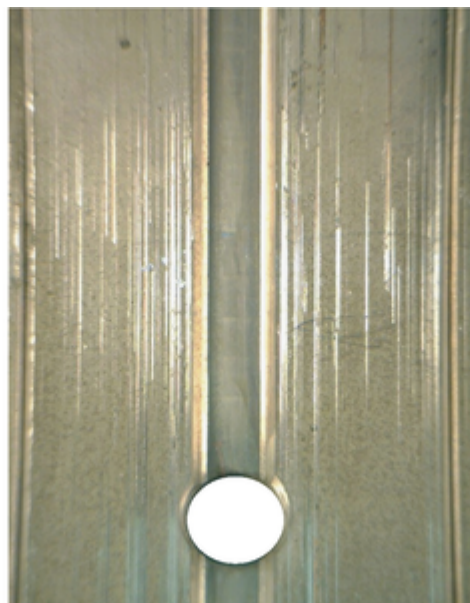
Figure 71: Trimetal lower main bearing from Figure 70 at 20 times magnification.

Figure 71 shows one of the trimetal lower main bearings from Figure 70 at 20 times magnification. It shows the most loaded area of the bearing shell. This is a trimetal bearing.

Note the damage that is the color of copper (1). This damage is termed large particle damage because the particle was large enough to reach through the lead/tin overlay, the nickel layer and into the bronze layer.

The marks (2) that are bright silver in color indicate small particle damage. If the damage can be felt with the fingernail it may be through to the nickel layer. If the damage can **not** be felt with the fingernail, then the particle was small enough to **only** mark the lead/tin overlay or flash.

The darker areas (3) are the lead/tin flash and overlay. The lighter areas indicate where a portion of the lead/tin flash and overlay have been worn away. The darker color is normal and is the result of interaction with chemicals in the oil.



01r00143

Figure 72: Trimetal upper main bearing from Figure 69 at 20 times magnification.

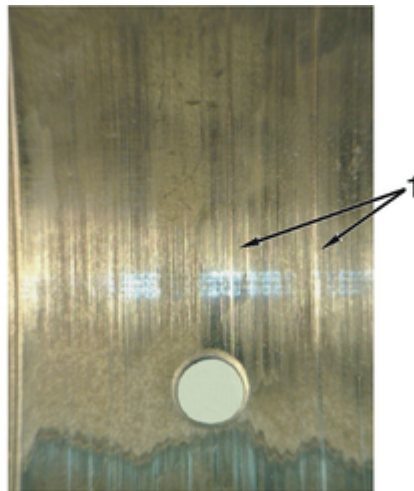
Figure 72 shows one of the trimetal upper main bearings from Figure 70 at 20 times magnification.

The damage that is copper in color at the edge of the oil groove is the result of a large particle passing between the crankshaft and the bearing. The particle was larger than the clearance between the crank and the bearing. The damage extends through the lead/tin flash and nickel layer into the bronze lining.

The marks that are shiny silver in color are the result of small particles passing between the crankshaft and the bearing. Note that some of these marks stop indicating there may still be a piece of debris in the embedded in the overlay material.

Note that there is less marking and damage on this upper bearing shell than on the lower bearing shell in Figure 71. This condition is typical and normal. The oil carrying the debris tends to stay in the groove in the bearing until it reaches the parting line of the upper and lower shells. Remember that the bearing shells are thinner near the parting line. The oil then tends to spread out as it flows to the lower shell. Also, there is less clearance between the crankshaft and the lower bearing due to the loading of the cylinders firing and the weight of the crankshaft plus the piston and connecting rod assemblies.

©Cummins Inc



01r00144

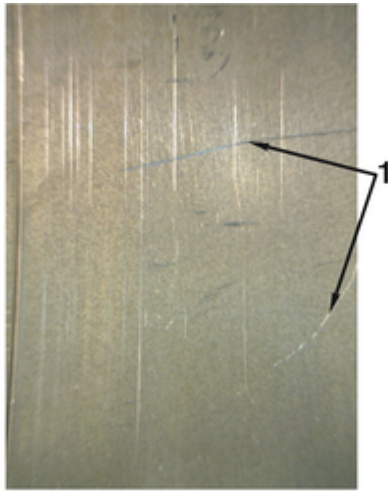
Figure 73: Trimetal upper rod bearing from Figure 69 at 20 times magnification.

Figure 73 shows one of the trimetal upper rod bearings from Figure 70 at 20 times magnification.

Two of the largest scratches (1) have barely penetrated the nickel layer of this trimetal bearing. This indicates that the particles that caused the scratches were larger than the clearance between the crankshaft and the bearing. The area shown in Figure 73 is the most heavily loaded portion of the bearing.

The other scratches seen have been made by smaller particles. Some of these scratches are through the lead/tin flash and overlay to the nickel layer. Some with the same appearance are **not** to the nickel overlay.

Note that there is less indication of debris on this bearing shell than on the lower main bearing shell. Typically more oil borne debris is found embedded in the main bearings than the rod bearings. The main bearings receive oil first and have a greater surface area so catch more of the debris.



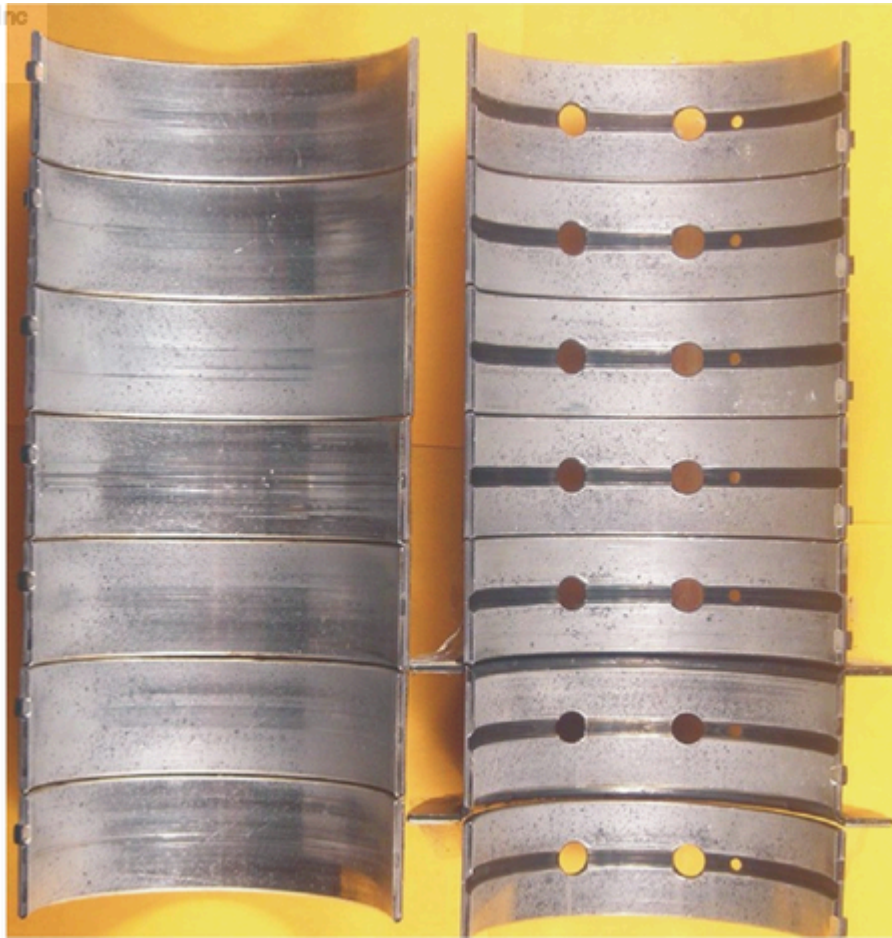
01r00145

Figure 74: Trimetal lower rod bearing from Figure 70 at 20 times magnification.

Figure 74 shows one of the trimetal lower rod bearings from Figure 70 at 20 times magnification.

Note : The marks (1) across the bearing in Figure 74 are the result of handling damage after removal from the engine. Engine operation will not leave marks across the bearing.

Note that there is very little evidence of debris in this bearing shell. That is because there is more clearance between the lower bearing and the crankshaft journal during engine operation. The upper bearing usually shows more debris damage than the upper bearing because of the lesser clearance at the upper bearing. Also, the oil film is thinner between the upper bearing and the crankshaft.



01r00146

Figure 75: Large particle debris damage to bimetal main bearings.

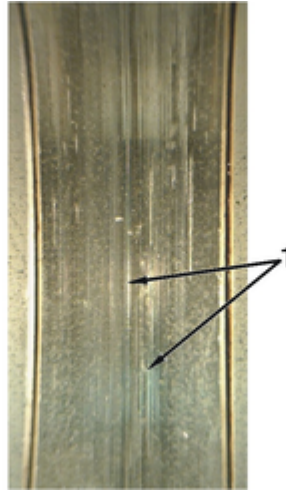
Figure 75 shows the upper and lower main bearings from a B6.7 engine. The engine suffered a piston skirt scuffing failure due to over-fueling at 96,561 kilometers [60,000 miles]. The pistons are made of aluminum.

This set of bearings exhibits large particle debris damage. While these bearing are **not** reusable, there is no failure on the part of the bearings. To the contrary, the bearings exhibit good embedability performance. The relatively soft particles of aluminum material have scratched the lining in some areas and the bearings have been able to withstand the debris.

The lubricating oil has carried the debris throughout the engine. Note that **all** of the bearings have some degree of debris damage. When **all** of the bearings indicate debris damage:

- The oil filter bypass valve has been pushed open during engine start (**all** oil filter bypass valves are opened until oil pressure is available to the clean oil side of the bypass valve).
- The oil filter has so much debris that the filter is plugged forcing the filter bypass valve open,
- The oil filter bypass valve is stuck open, possibly by the debris from the fuel pump failure.

See Figure 76 to Figure 78 for closer views of the bearing damage.



01r00147

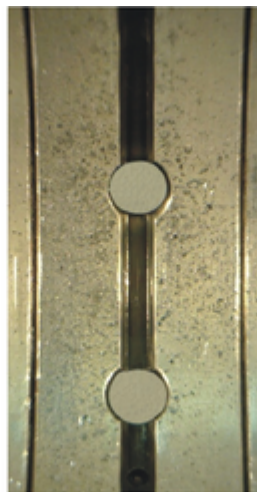
Figure 76: Bimetal lower center main bearing from Figure 75 at 20 times magnification.

Figure 76 is a photograph of the bimetal center lower main bearing from Figure 75 at 20 times magnification. The photograph is of the most loaded area of the bearing shell.

The damage is **not** as significant as the damage shown in the pictures of the trimetal bearing because the aluminum material from the piston is much softer than the ceramic material shown in the trimetal bearing.

The damage (1) is much deeper into the lining. This damage is termed large particle damage because the particle was large enough to leave a groove that is easily felt with the finger and fingernail.

The areas that are bright silver in color indicate small particle damage and some smearing of the lining material. These areas are usually **not** felt with the finger or fingernail.



01r00148

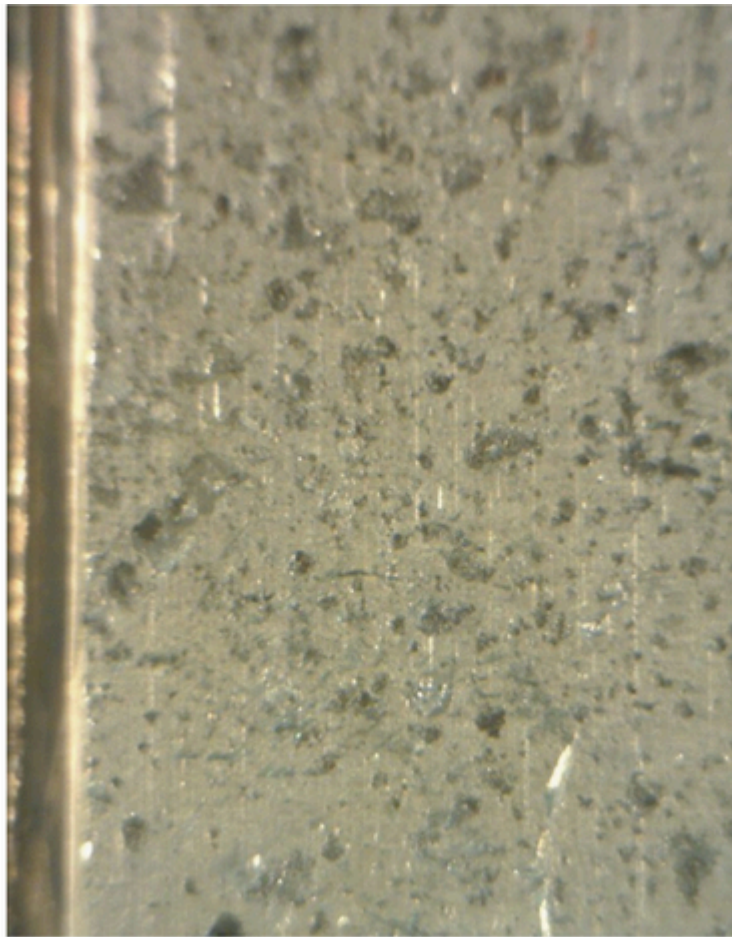
Figure 77: Bimetal upper center main bearing from Figure 75 at 20 times magnification.

Figure 77 is a photograph of the bimetal center upper main bearing from Figure 75 at 20 times magnification.

Note the absence of grooves or streaks from the debris. In this case the aluminum from the piston skirt is smaller than the clearance between the upper bearing and the shaft. The dots visible are actually small pieces of aluminum debris. The black dots are from the oil that did **not** wash completely away when the bearings were cleaned.

The material can be felt with the finger. Sometimes the material deposited is incorrectly determined to be pitting. The next photograph shows that what can be felt is **not** actually pitting.

©Cummins Inc



01r00149

Figure 78: Bimetal upper center main bearing from Figure 75 at 35 times magnification.

Figure 78 is a photograph of the bimetal center upper main bearing from Figure 75 at 35 times magnification. The intent of the photograph is to show that the debris from the piston failure and possibly the mating lower bearing is actually embedded in the bearing lining and **not** pitting.

Harmonic Lobing Break-In:

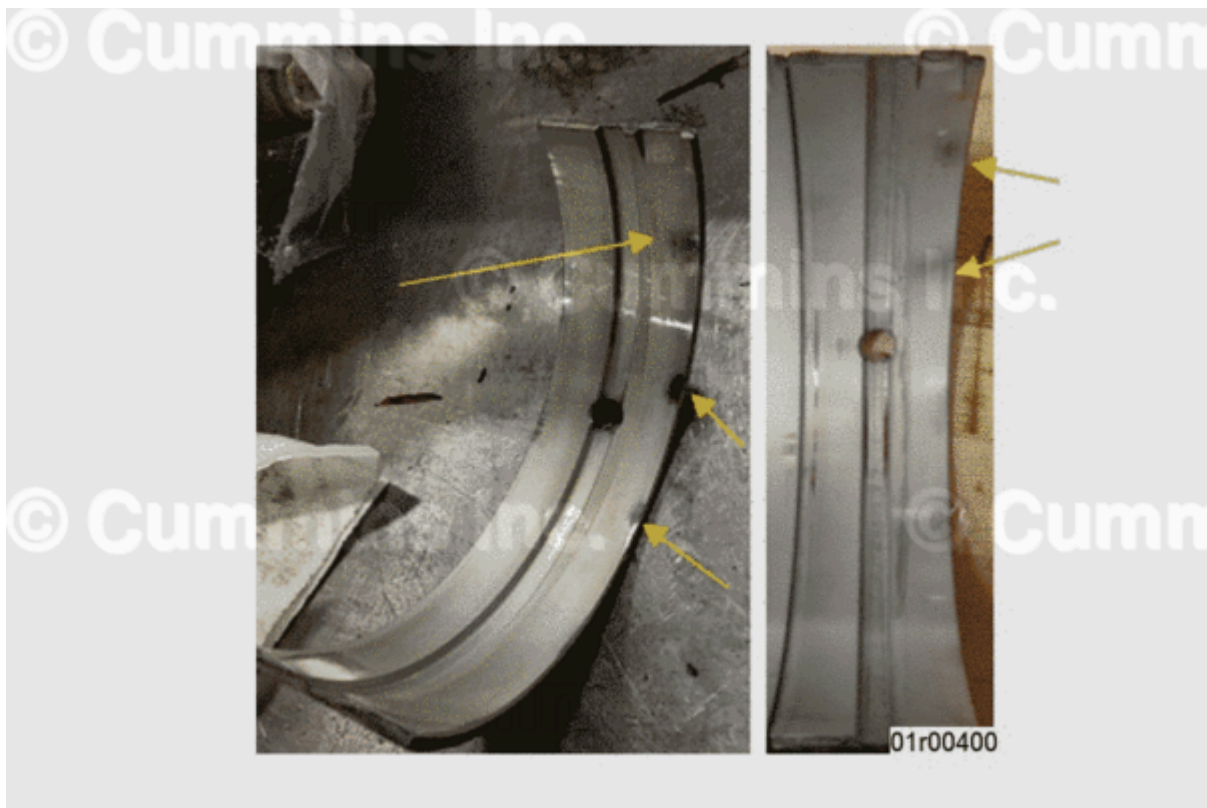


Figure 79: Upper (Cylinder Block Side) Trimetallic Main Bearing Examples with Harmonic Lobing Break-In Type Wear.

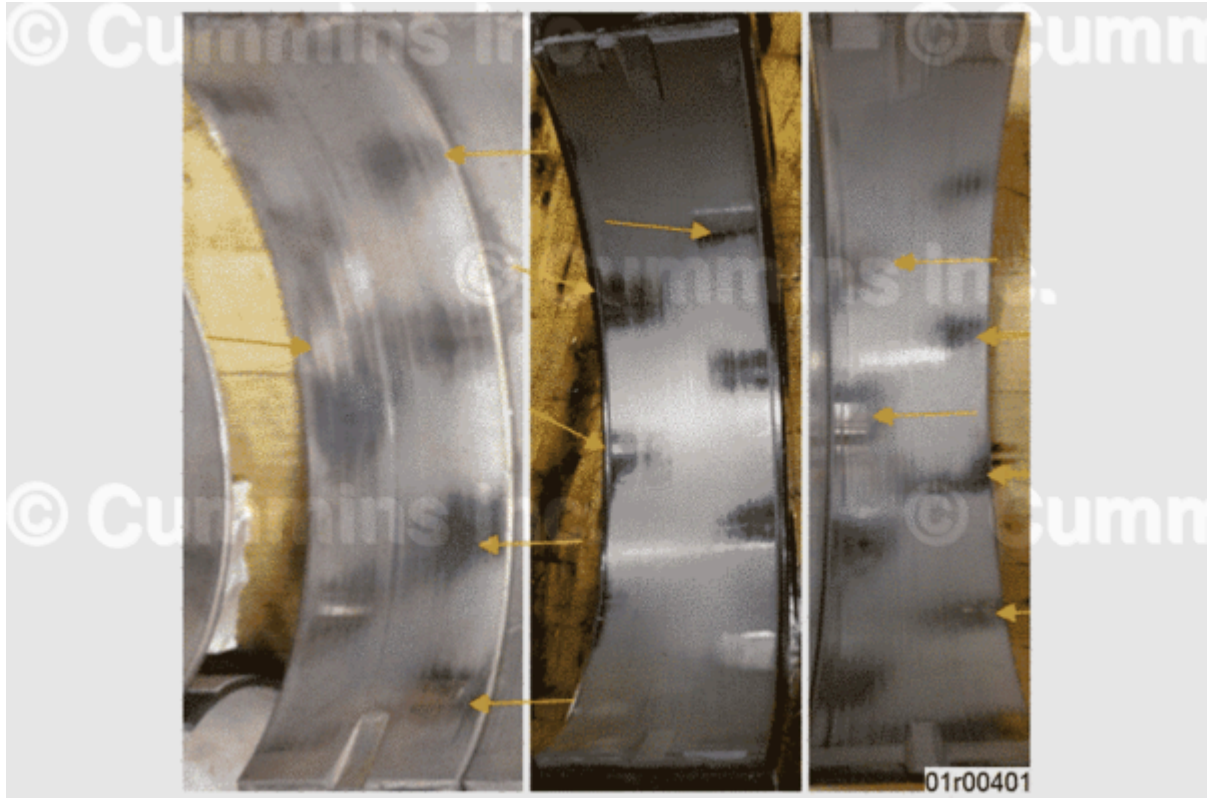


Figure 80: Lower (Cap Side) Trimetallic Main Bearing Examples with Harmonic Lobing Break-In Type Wear.

Figure 79 and 80 are photographs of wear in the tin flash and overlay layers. This amount of wear is acceptable and is due to a typical level of harmonic lobing break-in of the cylinder block main bearing bore. If any copper is visible, reference to the Loss of Overlay section above.

Prevention of Bearing and Bushing Failures

Most bearing damage and the resulting failure can be avoided by observing a few simple rules of caution during engine preparation and assembly, repair and operation and maintenance.

Refer to the General Cleaning instructions found on the front of **all** service manuals and/or troubleshooting and repair manuals.

Use the Clean Care Kit specified in repair procedures in **all** service manuals and/or troubleshooting and repair manuals.

Keep partially assembled engine and unit assemblies covered when **not** working on them. Make sure hand and special tools and lubricants are kept clean so as to no transfer dirt and debris to the engine.

Note : You can spend too much time cleaning parts, but you can not get the parts too clean.

Cleaning and Preparation – Cylinder Block:

Thoroughly clean **all** internal surfaces and oil passages. Removal of stuck gasket material **must not** remove or change the machining marks on the part. When removing composite head gaskets material, make sure the holes into the block and cylinder head are covered to keep material out of the engine interior. Use power tools to clean gasket surfaces very carefully so as to **not** disturb the machining marks. Disturbance of the machining marks removed metal that can fall into the engine and can damage the gasket surface that will allow a leak. **All** brands of abrasive pads contain aluminum oxide or similar material that is very hard. If material from the pads is left in the engine, either loose or embedded in parts from too much pressure on the tool, it can cause damage to bearings and other parts. **All** debris created by the use of abrasive materials **must** be completely removed. It is particularly important that dead-end passages in oil drillings be cleaned out. Remove **all** oil passage plugs. Often metal chips and other debris become trapped and embedded in the oil varnish in these locations. It is suggested that an “end” brush and/or a fly-type brush (**not** sandpaper) be used in the oil passages as they are flushed to break loose the sludge and wash it out. After cleaning, check oil passages with a piece of white cloth on a ramrod to make sure of adequate cleanliness. It is particularly important to pay attention to cleaning the oil passages and block surfaces after any machining (salvage) operations that remove material from the block. Use the appropriate chip catchers and protect the crankshaft when machining bores in a partially assembled engine. From a cleanliness standpoint, it is better that the cylinder block be completely disassembled if any machining is to be performed.

If any work is done to the main bore, check the alignment with proper service tools.

Cleaning and Preparation – Crankshaft:

Thoroughly clean **all** debris from the exterior and interior of the crankshaft. This is best done just before the crankshaft is to be installed in the cylinder block. New crankshafts are coated with a rust preventative that **must** be cleaned from the crankshaft before installation in the cylinder block. Brush and flush the oil passages to make sure they are clean and open. Some older crankshafts have pipe plugs in the oil drillings. These plugs **must** be removed to clean the oil drillings.

Cleaning the crankshaft while installed in the cylinder block can be difficult, but it can be done. Use a lint free cloth or paper and contact cleaner to clean the journals when the bearing shells are removed. If the connecting rods have been removed and the main are being inspected or replaced, use contact cleaner to spray into the oil drilling in the rod journal to flush any residual oil and dirt into and out of the oil drilling in the main bearing journal. It may be necessary to rotate the crankshaft to make sure the contact cleaner has a good drain path. Use clean compressed air to blow out the drilling.

Measure the main and rod journals if there is any doubt as to whether or **not** the crankshaft has been reconditioned. While reconditioning procedures call for marking the crankshaft to identify any undersize journals, that part of the process does **not** always happen.

Cleaning and Preparation – Connecting Rod:

Thoroughly clean the connecting rods. Brush and blow out any oil passages. Brush and blow out the threaded holes. **never** use a standard tap or die to repair or clean the threads in a connecting rod, rod cap or fastener. Many engines use fasteners with rolled threads and using a standard sharp-edged tap can cause a stress concentration that is likely to fail during engine operation. Check the fasteners and threaded holes by turning the fastener as far as possible through the threads by hand. There should be no tight spots or tools required to turn the fasteners.

Double check to be sure the correct rod and cap are installed together and in the direction specified in the service manual. Make sure the threads, the underside of the bolt head, any washer and nuts are properly lubricated with the oil specified. Make sure the calibration of the torque wrench used to torque the fasteners is up to date and correct.

Assembly – Main Bearings and Crankshaft:

If bearings are being reused, after an inspection as example, they **must** be used in the same bore with the same cylinder block and crankshaft. If either the block, crankshaft or bearing cap are changed, new upper and lower bearing shells **must** be used. **Never** replace just one of the bearing shells in a bore.

Make sure the bearing saddle in the cylinder block and the bearing surface of the main cap are clean. **Never** put lubricant on the steel back of a bearing to hold it in place. If the bearing shell will **not** stay in the cap or saddle, there is likely something wrong with the crush that **must** be corrected. Sometimes bearings that have been removed for inspection will fit more loosely in the cap than a new shell. In this case extra care is required to make sure the bearing stays in position.

If installing an upper main bearing shell with the crankshaft installed, use the proper service tool that fits in the oil drilling in the journal to push the shell into position. Be careful to **not** damage the parting line of the bearing shell or the crush may be reduced.

Lubricate the bearing shells as specified in the service manual. Make sure the lubricant used is **not** contaminated by dirt from the air or previous use. Keep the containers covered when the lubricant is **not** in use.

Wipe off **all** main bearing journals of the crankshaft before laying it in position on the upper main bearings. Be careful when installing the crankshaft to **not** move a bearing shell out of position or damage the crankshaft or the block. This is particularly true of those engines that have an integral upper main bearing and crankshaft thrust bearings. Usually the integral upper/thrust **must** be installed after the crankshaft is laying on **all** the upper main bearings. Great caution **must** be exercised to **not** damage the thrust bearing surfaces. Also be careful that the sharp corners of the main bearing saddles do **not** damage the crankshaft.

Lubricate and install the crankshaft thrust bearings if the engine is so equipped.

Install and lubricate the lower main bearing shells in the main caps. Lubricate the main cap bearing bolts as specified in the service manual. Place the bolts in the cap before placing the cap to the block. Hold the main cap off the saddle and start the threads on the main bolts by hand and turn the bolt two or three revolutions. Using a mallet to then drive the cap to the block at this point presents the risk of dislodging the bearing shell from the cap, particularly if the cylinder block is upside down. It is better to pull the cap to the block by alternately and evenly tightening the bolts to pull the cap to the block or used two ratchet/socket tools to turn the bolts simultaneously.

After **all** the main caps are installed and the main cap to block bolts are tightened to specification, turn the crankshaft two revolutions by hand to make sure there is no binding. If the crankshaft will **not** turn or a tool is required to turn the crankshaft (on engines 23 liters and less), make sure the problem is corrected.

On engines that have main cap side bolts, lubricate, install, and torque those bolts. Repeat turning the crankshaft two revolutions to make sure there is no binding.

Assembling Connecting Rods to Crankshaft:

If the bearings are being reused, after an inspection as example, they **must** be used in the same connecting rod on the same crankshaft journal with the same crankshaft. If either the rod or crankshaft are used, new upper and lower bearing shells **must** be used. Sometimes bearings that are reused are **not** as tight in the rod saddle as new parts. In this situation be very careful that the bearing does **not** fall out as the assembly is placed in the cylinder bore. It may be necessary to install the bearing after the piston and rod assembly are installed in the bore.

Install the upper rod bearing into its position in the rod saddle. Lubricate the bearing as specified in the shop manual. Use the appropriate guide tool specified in the shop manual to prevent damage to the bearing, rod, or crankshaft. If the guide tools are **not** available, obtain the help of another technician to guide the connecting rod to the crankshaft and make sure the bearing stays in the rod. Make sure the connecting rod and piston assembly are oriented toward the front of the engine as specified in the shop manual.

Install the lower rod bearing shell in the rod cap. Lubricate the bearing surface as specified in the shop manual. Lubricate the rods and fasteners as specified in the shop manual. Double check to be sure the correct rod and cap are being assembled together by comparing the markings on the parts.

Assemble the cap to the rod. It is better to use a hand tool to pull the rod cap to the rod than a power tool; the hand tool will prevent moving the bearing shell out of position.

After the rod fasteners are tightened to specification, manually shake the rod back and forth on the journal through the side clearance to make sure there is no binding. The connecting rod **must** be in the vertical position for this so the weight of the rod is **not** interfering with the feel for binding. Measure the rod side clearance as specified in the shop manual.

Document History

Date	Details
2015-11-2	Module Created
2017-3-14	none
2022-1-20	Quick Fix application:
2022-12-1	Addition of Harmonic Lobing Break-in wear pattern.